

Unit 12 Overview: Solving Multi-Step Problems With Proportional Relationships

Unit Narrative



This unit focuses on using proportional reasoning to solve problems, including those involving percentages. Students formally learn to set up and solve proportions. Students also explore how to calculate discounts, markups, taxes, interest, percent change, percent error, and simple interest. In Unit 5, students applied ratio relationships to solve real-world problems involving percentages.

In Section 1, students extend prior learning related to ratios and rates to formalize setting up and solving proportions to solve problems. Students revisit contexts addressed in prior units, such as equivalent ratios, scale factor, and scaling data to a population, to develop a stronger sense of proportional relationships. The unit then extends to include multi-step problems that require proportional reasoning with percentages to solve.

In Section 2, students apply their previous understanding of percentages and ratios from Units 3 and 5 to solve multi-step, real-world percentage problems. Students determine how to calculate percent change and apply their skills to determine taxes, tips, fees, markups, and discounts. Finally, students discover how to calculate percent error and simple interest.



Unit At-A-Glance

Section 1: Solving Problems with Proportions	Benchmarks Addressed	Lesson 1	Creating Proportions
	MA.7.AR.3.2	Lesson 2	Using Proportions to Solve Problems
		Lesson 3	Solving Real-World Problems Involving Proportions
Section 2: Using Percentages and Ratios to Solve Real-World Problems	Benchmarks Addressed	Lesson 4	Discounts and Markups
	MA.7.AR.3.1	Lesson 5	Determining Tax, Tip, or Fees Given a Percentage
		Lesson 6	Solving Multi-Step Problems Involving Discounts, Taxes, Tips, or Fees
		Lesson 7	Percent Increase and Decrease
		Lesson 8	Calculating Percent Error and Simple Interest



Differentiation

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Engagement The “WHY” of learning

- During 12.6.2 Exploration in lesson 6, consider creating a classroom store with items marked with prices, signs displaying discounts and markups, and a given sales tax. A classroom restaurant could also be created to practice calculating tax and tip in 12.5.4 Guided Instruction in lesson 5.

Partner A: Sweet & Salty

6.5 cups of popcorn
2 cups of frosted oat and marshmallow cereal
1.5 cups of mini pretzels
1 cup of chocolate chips
1 cup butterscotch chips

Partner B: Party Ranch

4 cups square wheat cereal
4 cups square rice cereal
2 cups mini pretzels
1.5 cups corn chips
1.5 cups cheddar crackers

Representation The “WHAT” of learning

- Utilize the multiple methods of solving proportions for students to navigate and identify what fluency and efficiency look and sound like for their unique learning needs. Each lesson in Section 1 provides a variety of methods and opportunities to examine student work.
 - Avoid language like “Which is the correct way to solve?” or “Who did it the right way?”
 - Embrace language like “Explain which method makes the most sense to you.” or “How was your approach the same or different than your partner’s? What does that make you think for the next time you might solve?”
- In 12.5.2 and 12.5.4 Guided Instruction components in lesson 5, prompt students to underline key words and annotate the examples in their own words. Debrief each component with a whole-class discussion to provide an auditory entry point.
- For 12.4.2 Guided Instruction in lesson 4, consider providing visual and kinesthetic entry points by having students look at sales flyers to see real-world examples of discounts.

A photo to poster $\frac{\text{height}}{\text{width}} \quad \frac{16 \text{ cm}}{10 \text{ cm}} = \frac{?}{25 \text{ cm}}$ $\frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5} = \frac{40 \text{ cm}}{25 \text{ cm}}$	B poster to photo $\frac{\text{width}}{\text{height}} \quad \frac{25 \text{ cm}}{25 \text{ cm}} = \frac{10 \text{ cm}}{?}$ $\frac{40 \text{ cm}}{25 \text{ cm}} = \frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5}$
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Action & Expression The “HOW” of learning

- Offer opportunities for students to use an audio recorder (such as Flip Grid) or speech-to-text software to explain their observations and learning as an alternative means of communicating their knowledge during Wrap-Ups where students are asked reflect or summarize in writing.

ELL Information

Consider creating a Frayer Model for each key term from the Glossary (found below):

- This graphic organizer helps students see multiple representations of key terms.
- This model will help students distinguish between the terms and formulas.
- Have students fill in the models as they work through the lessons. Ask students to share their models with the whole class so students can add helpful information they see their peers present.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

- Equation
- Proportional relationship

- Percent error
- Simple interest

Frayer Model

DEFINITION	CHARACTERISTICS																																										
A method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.	$I = P \times r \times t$ I = Interest P = Principal r = rate t = time																																										
Simple Interest																																											
EXAMPLES/MODELS	NON-EXAMPLES																																										
<table><tr><th colspan="3">Simple Interest</th></tr><tr><th>Years Invested</th><th>Calculation</th><th>Yearly Total (rounded to nearest whole)</th></tr><tr><td>1</td><td>$\\$1,000 \cdot 0.07 \cdot 1$</td><td>\$1,070</td></tr><tr><td>2</td><td>$\\$1,000 \cdot 0.07 \cdot 2$</td><td>\$1,140</td></tr><tr><td>3</td><td>$\\$1,000 \cdot 0.07 \cdot 3$</td><td>\$1,210</td></tr><tr><td>4</td><td>$\\$1,000 \cdot 0.07 \cdot 4$</td><td>\$1,280</td></tr><tr><td>5</td><td>$\\$1,000 \cdot 0.07 \cdot 5$</td><td>\$1,350</td></tr></table>	Simple Interest			Years Invested	Calculation	Yearly Total (rounded to nearest whole)	1	$\$1,000 \cdot 0.07 \cdot 1$	\$1,070	2	$\$1,000 \cdot 0.07 \cdot 2$	\$1,140	3	$\$1,000 \cdot 0.07 \cdot 3$	\$1,210	4	$\$1,000 \cdot 0.07 \cdot 4$	\$1,280	5	$\$1,000 \cdot 0.07 \cdot 5$	\$1,350	<table><tr><th colspan="3">Compound Interest</th></tr><tr><th>Year</th><th>Calculation</th><th>Yearly Total (rounded to the nearest whole)</th></tr><tr><td>1</td><td>$\\$1,000 (1.07)$</td><td>\$1,070</td></tr><tr><td>2</td><td>$\\$1,000 (1.07) (1.07)$</td><td>\$1,145</td></tr><tr><td>3</td><td>$\\$1,000 (1.07) (1.07) (1.07)$</td><td>\$1,225</td></tr><tr><td>4</td><td>$\\$1,000 (1.07) (1.07) (1.07) (1.07)$</td><td>\$1,311</td></tr><tr><td>5</td><td>$\\$1,000 (1.07) (1.07) (1.07) (1.07) (1.07)$</td><td>\$1,403</td></tr></table>	Compound Interest			Year	Calculation	Yearly Total (rounded to the nearest whole)	1	$\$1,000 (1.07)$	\$1,070	2	$\$1,000 (1.07) (1.07)$	\$1,145	3	$\$1,000 (1.07) (1.07) (1.07)$	\$1,225	4	$\$1,000 (1.07) (1.07) (1.07) (1.07)$	\$1,311	5	$\$1,000 (1.07) (1.07) (1.07) (1.07) (1.07)$	\$1,403
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- Percent of change
- Unit rate

Additional Differentiated Support

Scaffolding Resources

- a. If Xavier tips 20% on the subtotal \$24.95, how much tip will he pay?
\$4.99
- b. The city has a sales tax rate of 9.5%. How much in tax will the customer pay on the subtotal of \$24.95?
\$2.37
- c. What is the total Xavier spent on his meal, including tip and tax?
\$32.31

Date: Sep 19th	
Time: 6:04 PM	
Server: Flo	
Bread Stix	9.50
Ravioli Bites	10.50
Cheesecake	4.95
Subtotal	24.95
Sales Tax	—
Tip	—
Total	—

Mark Up Text

Throughout the unit, students will be solving real-world problems. This unit is text and reading heavy. When students translate words into numbers, algebraic expressions, and/or equations, encourage them to use a highlighter to highlight phrases and words to call attention to key terms. Using a small piece of text, teachers can model how to mark up before releasing the students to do the same. Students can choose to use annotations shared by the teacher or create their own key and markings to mark the text. Then they will turn to a partner to discuss each sentence and compare where they made marks. For the first few examples, have groups share their decisions with the class. Then guide students through turning the

marked-up text into an expression. ELL students may find this a helpful intermediary step, so encourage them to use this strategy even as you release them to work independently.

On Ramp

On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills:

- Real-World Addition/Subtraction Multi-Step Problems
- Real-World Multiplication/Division Multi-Step Problems
- Real-World Multiple Operations Multi-Step Problems
- Rounding Decimals to the Nearest Tenth
- Rounding Decimals to the Nearest Hundredth
- Adding Decimals
- Subtracting Decimals
- Multiplying Decimals
- Dividing Decimals
- Recognize and Generate Equivalent Fractions Using Models
- Evaluate Expressions
- Write and Interpret Expressions
- Adding Two-Digit and One-Digit Numbers
- Addition Fluency
- Subtraction Fluency



Section Overview

Section 1: Solving Problems with Proportions (Lessons 1-3)

Benchmarks Addressed	Learning Targets	
MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.	Lesson 1	• I can create proportions to represent a situation.
	Lesson 2	• I can solve problems using proportions.
	Lesson 3	• I can solve problems using proportions.

Section 2: Using Percentages and Ratios to Solve Real-World Problems (Lessons 4-8)

Benchmarks Addressed	Learning Targets	
MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems. <i>Clarification 1: Instruction includes discounts, markups, simple interest, tax, tips, fees, percent increase, percent decrease and percent error.</i>	Lesson 4	• I can solve problems involving discounts and markups.
	Lesson 5	• I can determine the amount of tax, tip, or fee given a percentage. • I can determine the percentage of a discount or markup.
	Lesson 6	• I can solve multi-step problems that involve discounts, markups, taxes, tips, or fees.
	Lesson 7	• I can find percent increase or percent decrease.
	Lesson 8	• I can find percent error. • I can solve problems that involve simple interest.



Vertical Alignment

Grade 6 Accelerated Unit 12
Solving Multi-Step Problems with
Proportional Relationships



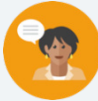
Grade 7 Accelerated Unit 7
Representing Proportional Relationships

Prior Course(s)	Course Level	Future Course(s)
	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.	MA.7.AR.4.1 Determine whether two quantities have a proportional relationship by examining a table, graph, or written description. MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship. MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, an equation, or a written description. MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation. MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

12.1 Creating Proportions

Lesson Introduction

	Benchmark	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.			Learning Target	I can create proportions to represent a situation.
	Benchmark Coherence	Building On MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.		Working Toward MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.		
	Essential Questions	- How do you use proportions to represent a situation and find missing information? - What role does unit rate play in proportional relationships?				
	Lesson Narrative	This lesson formally introduces creating proportions to represent situations, focusing on how multiple proportions can represent a single situation and that solving proportions reveals different information about the initial problem. This lesson deeply dives into a single-problem context to explore the different ways proportions can be set up, moving slowly from procedures to concepts before releasing students to a different context in the 12.1.6 Collaboration. The contexts used in this lesson spiral content from previous units to expose students to additional proportional reasoning about those contexts.				
	Component Summary	12.1.1 Warm-Up (~5 min.)	Consider a Math Flop (<i>SE p. 213</i>)	12.1.4 Guided Instruction (5 min.)	Study of the structure of proportions using length units (<i>SE p. 215</i>)	
		12.1.2 Exploration (5-10 min.)	Consider solution strategies in estimating with scale factors (<i>SE p. 214</i>)	12.1.5 Guided Practice (5-10 min.)	Encounter contextualized proportion building with whole numbers (<i>SE p. 216</i>)	
		12.1.3 Bring It Together (5 min.)	Define “proportional relationship,” “proportion,” and “unit rate” (<i>SE p. 215</i>)	12.1.6 Collaboration (10-15 min.)	Apply methods of solving with proportions (<i>SE p. 216</i>)	
		12.1.7 Wrap-Up (~5 min.)	Analyze the proportions of three students to find the errors (<i>SE p. 217</i>)			
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - Proportional relationship - Proportion - Unit rate Additional Terms: N/A		N/A		N/A

 Teacher Tips	Common Misconceptions - Students may struggle to interpret proportions that are structured in a way that is different from their original thinking. - Students may focus on one quantity instead of the relationship between quantities. - Students may confuse the glossary terms of rate, ratio, and proportion. - Students may try to solve proportions using additive reasoning rather than multiplicative thinking.	Things to Consider - Instruction should avoid referencing a “correct” way to set up proportions and instead highlight multiple pathways. - Students may attempt to use other incorrect proportions as a check for whether another incorrect proportion is viable or not. Both will result in an incorrect value, so the focus for this lesson should be on ensuring the ratios are correct before calculating. - Students should be encouraged to label the ratios they use in their proportions to ensure they are setting them up in a way that will result in accurate, equivalent ratios.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Utilize a Think-Pair-Share for 12.1.2 Exploration questions 2 and 3. Give students time to make observations and find the similarities and differences independently before sharing their observations with a partner.	MA.K12.MTR.2.1: Multiple Representations Students will be working to represent the relationships between values within a problem context. Starting in 12.1.2 Exploration, highlight the process by which students are moving from understanding connections between their comprehension of the problem and representing that as a unit rate versus with equivalent ratios in 12.1.4 Guided Instruction and then the connection to creating a proportion in 12.1.6 Collaboration.
 Differentiate Instruction	Flexible pathways for problem-solving provide a variety of entry points for students. 12.1.4 Guided Instruction naturally differentiates instruction for students in providing four different methods for proportions, giving students a choice in which method makes the most sense to them. Make space for student choices in methods for problem-solving.	
Lesson Notes:		

12.1.1 Warm-Up: Math Flop

Component Overview: Students make observations on math in a real-world context, specifically targeting the error in algebraic and visual representations.



12.1.1 Warm-Up: Math Flop

The advertisement encourages making DiGiorno pizza a part of your weekly meals.



For students who might not notice the Math Flop:

- How much is one portion?
- What is the visual representation of the pizza trying to communicate?
- Which is more: $\frac{1}{5}$ or $\frac{1}{4}$?

1. What do you notice about the advertisement?

Answers will vary. It says one portion of the pizza is $\frac{1}{5}$ but the image shows fourths.

12.1.2 Exploration: Proportional Thinking

Component Overview: Students consider the measuring tools and processes a student uses to estimate a missing height of an enlarged poster from a photograph. Then students analyze two different ways students used to calculate the missing height measurement using proportional reasoning. From these examples, students describe the similarities and differences between the calculation strategies.



12.1.2 Exploration: Proportional Thinking

A photograph is enlarged to make a poster. Measurements are given in centimeters (cm).

16 cm



10 cm

?



25 cm

Abigail used her pencil and measured the photo's width using her finger to mark off 10 cm on the pencil. She then used the pencil to estimate the height using the mark on her pencil.

1. Discuss with your partner why Abigail's strategy could lead to an incorrect answer.

Answers may vary. Abigail's strategy may be useful for estimating, but not for accurate or precise measurements. Abigail did not use a ruler or equivalent tool to measure the photograph.



The Exploration purposefully creates the need for a proportion before trying the calculations. Monitor for student responses that reflect proportional reasoning to share with the whole group.

For example, students might say, "Abigail could do that, but her answer might be a little bit off because her measurement will not be as precise."

Would Abigail's strategy work for finding the distance between two cities on a map? Why or why not?

12.1.2 Exploration: Proportional Thinking (continued)

Mariana and Yousef used different strategies to find the height of the poster. Review their solution strategies. Then, work with your partner to answer the questions that follow.

Mariana's Work	Yousef's Work
$\frac{25 \text{ cm}}{10 \text{ cm}} \div \frac{10}{10} = \frac{2.5}{1} = 2.5$	$\frac{\text{height}}{\text{width}} \quad \frac{16 \text{ cm}}{10 \text{ cm}} = \frac{?}{25 \text{ cm}}$
Scale factor = 2.5 $16 \text{ cm} \times 2.5 = 40 \text{ cm}$	$\frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5} = \frac{40 \text{ cm}}{25 \text{ cm}}$
The height of the poster is 40 cm.	The height of the poster is 40 cm.

2. How are Mariana and Yousef's solution strategies similar?

*Actual student answers will vary.
Both multiplied 16 by 2.5 to find 40 cm.
Both got the same answer.*

3. How are their solution strategies different?

*Actual student answers will vary.
Mariana found a unit rate (scale factor).
Yousef set up a proportion.*



Think-Pair-Share

Sample routine strategy:

- *Think:* Take two minutes to make observations about Mariana's work and Yousef's work. What is the same? What is different? Make at least two observations for what is the same and at least two observations for what is different. There are no wrong answers—the more creative, the better!
- *Pair:* Turn to your shoulder partner and share your similarities and differences. Place a check mark next to any that are the same and circle the ones that are different.
- *Share:* Share one similarity or difference that your partner had that was unique.

Monitor for student responses in the Pair that highlight learning targets.

12.1.3 Bring It Together: Proportional Relationships

Component Overview: Students are introduced to the definitions for proportional relationships, proportions, and unit rates. Students then refer to these definitions and the examples from the previous activity to complete a written description in paragraph form of the two students' work.



12.1.3 Bring it Together: Proportional Relationships



In a **proportional relationship**, the values for one quantity are each multiplied by the same number to get the values for the other quantity.

A **proportion** sets two or more equivalent ratios in such relationships as equal to each other and can be used to solve problems.



A **unit rate** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

1. Use the words and phrases in the bank below to complete the description of Mariana and Yousef's work.

is equal to proportion ratio unit rate

The photograph and the poster are in a **proportional relationship** because the poster is a scaled version of the photograph. Mariana found the scale factor, which is a unit rate, and multiplied it by the height of the photo to find the height of the poster. Yousef set up a proportion stating the ratio of the height to width of the photograph is equal to the ratio of the height to width of the poster.



Consider having students notate and make connections between these terms and Mariana's and Yousef's work in the previous activity, as shown in the key.



How is the concept of proportions similar to or different from what you have done in previous lessons with:

- money?
- weight and mass?
- volume?

Consider referencing Abigail's work in the 12.1.2 Exploration. What would you tell Abigail now?

12.1.4 Guided Instruction: Proportional Thinking

Component Overview: Within the same problem context, students encounter four different strategies to create a proportion that can be used to find the missing enlarged height. Students then identify from these completed examples the scale factor between the photograph and the poster as well as the scale factor between the height and the width.



12.1.4 Guided Instruction: Proportional Thinking

Proportions can be set up in different ways using the relationships between each quantity.

Four different proportions that can be used to find the height of the poster are shown.

A photo to poster $\frac{\text{height}}{\text{width}} \left \frac{16 \text{ cm}}{10 \text{ cm}} = \frac{?}{25 \text{ cm}} \right.$ $\frac{16 \text{ cm}}{10 \text{ cm}} \times \frac{2.5}{2.5} = \frac{40 \text{ cm}}{25 \text{ cm}}$	B poster to photo $\frac{\text{width}}{\text{height}} \left \frac{25 \text{ cm}}{?} = \frac{10 \text{ cm}}{16 \text{ cm}} \right.$ $\frac{25 \text{ cm}}{40 \text{ cm}} = \frac{10 \text{ cm}}{16 \text{ cm}} \times \frac{2.5}{2.5}$
C width to height $\frac{\text{poster}}{\text{photo}} \left \frac{10 \text{ cm}}{25 \text{ cm}} = \frac{16 \text{ cm}}{?} \right.$ $\frac{10 \text{ cm}}{25 \text{ cm}} \times \frac{1.6}{1.6} = \frac{16 \text{ cm}}{40 \text{ cm}}$	D height to width $\frac{\text{photo}}{\text{poster}} \left \frac{?}{16 \text{ cm}} = \frac{25 \text{ cm}}{10 \text{ cm}} \right.$ $\frac{40 \text{ cm}}{16 \text{ cm}} = \frac{25 \text{ cm}}{10 \text{ cm}} \times \frac{1.6}{1.6}$

All of the proportions show that the ratios of the dimensions of the photograph and the dimensions of the poster are equivalent.

Complete the statements based on the proportions.

- The dimensions of the poster are ☐ 1.6 ☐ 2.5 times larger than the photograph.
- The height of both the photograph and poster is ☐ 1.6 ☐ 2.5 times larger than the width.



Number Talk

Consider a Number Talk routine to have students notice the similarities and differences between each proportion, ultimately concluding that all proportions can be used (flexible pathways).

Compare and contrast each method. Make note of three things that are the same and one thing that is different or unique to each method.



Consider Four Corners for a kinesthetic activity, with students discussing at each corner which proportion makes the most sense to them.

12.1.5 Guided Practice: Proportional Thinking

Component Overview: In this component, students are provided with a new context, moving from a sample to a population using proportions. Starting with identifying the proportional relationship between the households (with dogs and total) and sample to population. Students then compare two proportions, correcting the incorrect one and choosing one to use to solve.



12.1.5 Guided Practice: Proportional Thinking

The city of Tampa collected sample data on pet ownership to use when determining whether or not it needed to build additional dog parks. A city parks officer randomly sampled 200 households in Tampa and found that 126 of the households had a dog. Using the results of the sample, determine how many households have a dog if there are 153,918 households in Tampa.

- Complete the statements to describe the relationships that are being compared.
 - The number of households with a dog is being compared to the total number of households.
 - The data sample is being compared to the population of Tampa.
- Two proportions that represent this situation are below, but one of them has an error.

Proportion A		Proportion B	
$\frac{\text{Sample Dogs}}{\text{Total}} = \frac{126}{200}$	$= \frac{153,918}{x}$	$\frac{\text{Total Population}}{\text{Sample}} = \frac{153,918}{200}$	$= \frac{x}{126}$

Proportion ☐ A ☐ B is not set up correctly. The proportion should be $\frac{126}{200} = \frac{x}{153,918}$.

- Find the number of households expected to have a dog.
≈ 96,968 households
Teachers can ask students why it makes sense to round the households to the nearest whole number. Potential answer: Households involve people and are not measured in fractions or parts.



Notice the embedded scaffolding in questions 1-3.



How else could this proportion have been correctly set up? (Using similar thinking to the previous component.)



For question 3, have students discuss why we needed to round the number of households.

12.1.6 Collaboration: Creating Proportions

Component Overview: Students continue to write two possible proportions given the problem context and pick one to use in the context of speed and a scale drawing.



12.1.6 Collaboration: Creating Proportions

1. Javad drove his car to Canada. He was driving on a road and saw the sign shown to the right. According to his speedometer, Javad was driving 75 miles per hour when he was pulled over for speeding. Canadian speed limits are given in kilometers (km) per hour because they use the metric system. There are approximately 1.61 kilometers in 1 mile.



- a. Create two different proportions that could be used to find Javad's speed in kilometers per hour.

Answers may vary.

Proportion A	Proportion B
$\frac{1.61 \text{ km}}{1 \text{ mi}} = \frac{x \text{ kmph}}{75 \text{ mph}}$	$\frac{x}{1.61 \text{ km}} = \frac{75 \text{ mph}}{1 \text{ mi}}$

- b. How fast was Javad traveling in kilometers per hour?

$$\frac{75 \cancel{\text{mi}}}{1 \text{ hr}} \times \frac{1.61 \text{ km}}{1 \cancel{\text{mi}}} =$$

$$120.75 \text{ km/h}$$



Notice the contexts used in this lesson are familiar contexts from previous lessons or units, including scale drawings and using data to draw conclusions about populations.

12.1.6 Collaboration: Creating Proportions (continued)

2. An architect made a scale drawing of a house as shown.



The scale for the drawing is $\frac{1}{4}$ inch = 6 feet.

- a. Create two different proportions that could be used to find the actual length of the house represented by the drawing.

Answers will vary.

Proportion A	Proportion B
$\frac{\frac{1}{4} \text{ in.}}{6 \text{ ft.}} = \frac{3.5 \text{ in.}}{x}$	$\frac{3.5 \text{ in.}}{\frac{1}{4} \text{ in.}} = \frac{x}{6 \text{ ft.}}$

- b. What is the length of the house represented by the drawing?

84 ft.



Consider prompting students while monitoring to note rounding precision:

- How does the context inform how you are rounding?
- Would this be the same if there was no context, only calculations? Why or why not?

12.1.7 Wrap-Up: Error Analysis

Component Overview: This formative assessment helps you identify whether students can relate the context of the problem to the structure of the proportion. Use this data to inform which questions and supports you should emphasize in the next lesson.



12.1.7 Wrap-Up: Error Analysis

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. = 90 miles (mi).

Three students set up proportions to find the actual distance between Jacksonville and Gainesville, but two of them made errors.

- For each proportion, check whether it is correct or incorrect. If it is incorrect, fix the error by changing one of the ratios in the proportion.

Lily's Proportion	Eli's Proportion	Mia's Proportion
$\frac{2}{1.4} = \frac{90}{1.4x}$	$\frac{1.4}{2} = \frac{90}{x}$	$\frac{1.4}{x} = \frac{2}{90}$
☐ correct ☒ incorrect	☐ correct ☒ incorrect	☒ correct ☐ incorrect



For homework, consider having students to write an explanation of what they changed and why as if they were explaining it to a friend using the vocabulary words from 12.1.3 Bring It Together.

12.2 Using Proportions to Solve Problems

Lesson Introduction

 Benchmark	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.		 Learning Target	I can solve problems using proportions.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.		MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.	
 Essential Questions	- How can proportional reasoning help me find unknowns in everyday life? - How does scaling ratios help when finding missing parts of equivalent ratios?			
 Lesson Narrative	This lesson extends previous learning to focus on two specific ways to solve a proportion (multiplying both parts of a ratio by the same factor or solving the equation for the unknown value). Students then examine various students' work that highlight specific misconceptions/errors students make before they try solving problems using proportions on their own.			
 Component Summary	12.2.1 Warm-Up (~5 min.)	Using proportions for a recipe (<i>SE p. 218</i>)	12.2.4 Bring It Together (5 min.)	Checkpoint for student reflection (<i>SE p. 222</i>)
	12.2.2 Guided Instruction (10 min.)	Comparing two methods for solving proportions (<i>SE p. 219</i>)	12.2.5 Your Turn! (5-10 min.)	Solving three problems, creating and then solving proportions with no rounding (<i>SE p. 223</i>)
	12.2.3 Collaboration (15-20 min.)	Small group error analysis with four examples of misconceptions (<i>SE p. 220</i>)	12.2.6 Wrap-Up (~5 min.)	Reflection on a number line (<i>SE p. 223</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Scale factor		N/A	12.2.3 Collaboration could be done as a Gallery Walk. Consider copying or printing the prompts from the four student problems onto chart paper. For smaller groups, make two copies of each poster for a total of eight posters for approximately three to four students per poster.

 Teacher Tips	Common Misconceptions		Things to Consider
	- Students may look for the “correct” or “right way” to set up proportions and then incorrectly identify other methods of proportions as “incorrect.” - Students may try to add or subtract instead of multiply or divide when solving proportions. - Students may multiply numerator times numerator and denominator times denominator rather than correctly solving a proportion.		Instruction should not include cross-multiplying as a strategy. Cross-multiplication does not build students’ understanding and often creates a misconception for multiplying fractions.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard	
 Differentiate Instruction	Clarify, Critique, and Connect Activities 12.2.2 Guided Instruction and 12.2.3 Collaboration allow students the opportunity to correct their errors and clarify their meaning in the context of others to strengthen their problem-solving and calculations. Use these activities as a space where you facilitate the discussion to uncover student thinking.	MA.K12.MTR.4.1: Discussion and Reflection 12.2.3 Collaboration provides an opportunity for students to engage in mathematical discussion by analyzing and critiquing the thinking of others. Note the emphasis in <i>process</i> over <i>product</i> as students analyze the thinking and work of others more than the answer.	
	Consider a Gallery Walk for 12.2.3 Collaboration. Students can use strategies like Mark-Up for visual scaffolding in identifying key misconceptions or displayed student work. Foster discussion with specific prompts using the questions on each poster in the activity or with specific roles within each group. For example, one student can be the scribe, tasked with taking notes on what students are learning at each poster, while another student is the host, tasked with making sure student talk is equitable. The posters around the room provide opportunities for movement.		
Lesson Notes:			

12.2.1 Warm-Up: Making More Servings

Component Overview: Students encounter a recipe that needs to be scaled up. Though this includes scale factors, that word doesn't need to be defined yet in the lesson. This is similar to what was done in the previous lesson.



12.2.1 Warm-Up: Making More Servings

- Gabriel wants to make 5 servings of oatmeal for breakfast using this recipe. How much water and oats will he need to use? Show your work.

TRADITIONAL STEEL CUT OATS

Hot Water or Milk Preparation

1 Serving

- 1-1/2 cups water or milk $\times 5$
- 1/4 cup Quaker Steel Cut Oats $\times 5$

Directions:

1. Bring water or milk to a boil in a medium saucepan.
2. Stir in oats, reduce heat to low.
3. Simmer uncovered over low heat, stirring occasionally, for 25-30 minutes or until oats are of desired texture.

7.5 cups of water

1.25 cups of oats



Consider this spiraled practice and not additional instruction. Monitor for student work both to monitor progress from the previous lesson and to use as observational data for small groups in 12.2.3 Collaboration.

12.2.2 Guided Instruction: Solving Proportions

Component Overview: Students will encounter two methods of solving a proportion in two different contexts.



12.2.2 Guided Instruction: Solving Proportions

Just as there are different ways to set up a proportion to solve a problem, there are different ways to solve the proportion once it's set up.

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. = 90 miles. The proportion below can be used to find the actual distance between Jacksonville and Gainesville.

$$\frac{2}{90} = \frac{1.4}{x}$$

Two methods for solving the proportion are shown.

Method A Using a Scale Factor	Method B Solving the Equation
$\frac{2}{90} = \frac{1.4}{x}$ Determine the factor multiplied by 2 to get 1.4. $1.4 \div 2 = 0.7$ $\frac{2}{90} \times \frac{0.7}{0.7} = \frac{1.4}{63}$	$\frac{2}{90} = \frac{1.4}{x}$ $\cancel{90} \cdot \frac{2}{\cancel{90}} = \frac{1.4}{x} \cdot 90$ $x \cdot 2 = \frac{126}{\cancel{x}} \cdot \cancel{x}$ $\frac{2x}{2} = \frac{126}{2}$ $x = 63$
Both solution methods show the approximate distance between Jacksonville and Gainesville is 63 miles.	



- How is this the same as what you did in the previous lesson? How is it different? (Both methods got the same answer.)
- Use these questions to continue highlighting for students the flexible pathways for constructing and solving proportions.

- In Method A, how does multiplying the ratio $\frac{2}{90}$ by $\frac{0.7}{0.7}$

show that $\frac{2}{90}$ and $\frac{1.4}{63}$ are equivalent?

When both the numerator and denominator are multiplied by the same value, the value remains equivalent.

12.2.2 Guided Instruction: Solving Proportions (continued)

2. In Method B, why were both ratios multiplied by 90 and then by x ?

To balance the equation and maintain equivalence.

3. In Method B, why are both sides of the equation multiplied or divided by the same value in each step?

To balance the equation and maintain equivalence.

A certain shade of light blue paint is made by mixing $2\frac{1}{2}$ quarts of blue paint with 7 quarts of white paint. To make the same shade of light blue, how much white paint would be needed to mix with 4 quarts of blue paint?

4. Complete the proportion using the information given.

$$\frac{2.5}{4} = \frac{7}{x}$$

5. With your partner, decide who is Partner A and who is Partner B. Then solve the proportion using the method indicated in the space provided.

Partner A - Method A Using a Scale Factor	Partner B - Method B Solving the Equation
$\frac{2.5}{4} = \frac{7}{x}$ $7 \div 2.5 = 2.8$ $4 \times 2.8 = 11.2 \text{ qt.}$	$\cancel{4} \times \frac{2.5}{4} = \frac{7}{x} \times 4$ $x \times 2.5 = \frac{2.8}{\cancel{4}} \times \cancel{4}$ $\frac{\cancel{2.5}x}{\cancel{2.5}} = \frac{28}{2.5} \quad x = 11.2 \text{ qt.}$

6. Check your work with your partner, making any necessary corrections. Once both partners agree, write your partner's method in your workbook. Then complete the statement.

To make the same shade of light blue, 11.2 quarts of white paint would be needed.



Consider prompting students to find the distance between their hometown and the city of a family member using a map. This can be a project-based extension at the end of the unit or homework.



Clarify, Critique, and Connect

Consider constructing questions 4-6 as a language routine, with the teacher acting as the facilitator critique. Students complete question 4 before clarifying with the whole group. Students then complete question 5 before critiquing and connecting their methods in question 6. The teacher should monitor student responses during question 6 to surface accurate mathematical thinking or misconceptions whole group.

12.2.3 Collaboration: Examining Student Work

Component Overview: Students encounter four different sets of student work for four different problems to identify what the sample student work did correctly and incorrectly and then create a correct proportion and solve.



12.2.3 Collaboration: Examining Student Work

Four students solved different problems involving proportional relationships. Each student's work is shown, including any errors they may have made. Work with your partner or a small group to examine each student's work and complete the table that follows.

Student	Javier
Problem	A mixture of concrete is made up of sand and cement in a ratio of 5:3. How many cubic feet of sand is needed to make 160 cubic feet of concrete?
Student's Work	$\frac{5}{3} = \frac{160}{x}$ $x = 96 \text{ ft}^3$

What did Javier do correctly?	What error(s) did Javier make?	Find the correct answer.
He solved the proportion correctly.	He set up the proportion wrong. He used sand to cement instead of sand to concrete (total parts).	$\frac{5}{8} = \frac{x}{160}$ $x = 100 \text{ ft}^3$



Consider a variety of methods for facilitating collaboration based on your students and classroom.

Gallery Walk

- Round 1: Have students answer only the first question about what the student did correctly. Give them five minutes to discuss and write their answer on the poster.
- Round 2: Groups move to the next poster to answer the second question within five minutes.
- Round 3: Groups move to the next poster to answer the third question within five minutes.
- Round 4: Groups move to their final poster to review and revise the thoughts of all the other groups for all three questions.



Teacher Note for Javier's Work

Students should be noting the difference between part: part (5:3) and part: whole (5:8), with Javier incorrectly using 160 as part instead of whole.

12.2.3 Collaboration: Examining Student Work (continued)

Student	Kayla		
Problem	A recipe for making 4 pancakes calls for the ingredients listed. □ 6 tablespoons flour □ $\frac{1}{4}$ pint milk □ 1 pinch salt □ 1 egg □ $\frac{1}{3}$ pint water You want to make 10 pancakes. a. How much flour do you need? b. How much milk do you need?		
Student's Work	a. How much flour do you need? $\begin{array}{l} 10 : 4 = ? : 6 \\ \text{Flour} = \frac{10}{4} = \frac{5}{2} \\ \text{10 pancakes} = 6 + 6 + 3 \\ = 15 \text{ spoonfuls of flour} \end{array}$ b. How much milk do you need? $10 \text{ pancakes} = \frac{1}{4} + \frac{1}{4} + \frac{1}{8} = \frac{3}{8} \text{ pints}$		
What did Kayla do correctly?	What error(s) did Kayla make?	Find the correct answers.	
• She found the right amount of flour needed. • She used a strategy that would work by relating what is needed for 4 to 10 pancakes.	• She used the wrong units for flour (she used spoonfuls instead of tablespoons). • She added fractions incorrectly. (Find common denominator first.)	15 Tbsp $\frac{2}{8} + \frac{2}{8} + \frac{1}{8}$ $= \frac{5}{8} \text{ pt. milk}$	



Teacher Note for Kayla's Work

Students may note that she set up the proportion "incorrectly." Prompt students to recognize the variety of methods from the previous lesson and how different methods can still produce the same result.

Examples:

$\frac{\text{pancake (10)}}{\text{pancake (4)}} = \frac{\text{flour (x)}}{\text{flour (6)}}$ or $\frac{\text{pancake (4)}}{\text{flour (6)}} = \frac{\text{pancake (10)}}{\text{flour (x)}}$
 look different, but produce the same result.

12.2.3 Collaboration: Examining Student Work (continued)

Student	Isaac
Problem	A drawing of a surfboard in a catalog shows its length as $8\frac{4}{9}$ inches (in.). Find the actual length of the surfboard if $\frac{1}{2}$ in. length on the drawing corresponds to $\frac{3}{8}$ foot of actual length.
Student's Work	$\frac{\frac{1}{2}}{8\frac{4}{9}} = \frac{\frac{3}{8}}{s}$ $s \cdot 0.5 = \frac{0.38}{5} \cdot 8.44$ $s \cdot 0.5 = 3.2072$ $\frac{0.5s}{0.5} = \frac{3.2072}{0.5}$ $s = 6.4 \text{ in.}$

What did Isaac do correctly?	What error(s) did Isaac make?	Find the correct answers.
- He set up the proportion correctly. - He used a correct strategy to solve it.	He rounded the fractions, so he ended up with an inaccurate answer.	$\frac{\frac{1}{2}}{8\frac{4}{9}} = \frac{\frac{3}{8}}{s}$ $s = \frac{76}{9} \times \frac{3}{8}$ $s = \frac{228}{72} = \frac{228}{72} \times \frac{2}{1}$ $s = \frac{456}{72} = \frac{19}{3}$ $s = 6\frac{1}{3}$



Teacher Note for Isaac's Work
 Fluency with fractions will be a helpful skill in completing Isaac's work.

12.2.3 Collaboration: Examining Student Work (continued)

Student	Mari		
Problem	Juan made 13 out of 20 free throws at basketball practice. If Jada shoots 25 free throws, what is the smallest number of shots she has to make to have a better free-throw percentage than Juan?		
Student's Work	$\frac{13}{20} = \frac{f}{25}$ $20 + 5 = 25$ $13 + 5 = 18$ Jada has to make 18 shots.		
What did Mari do correctly?	What error(s) did Mari make?	Find the correct answers.	
She set up the proportion correctly.	She added instead of multiplying to solve the equation.	$\frac{13}{20} = \frac{f}{25}$ $\times 1.25$ $\times 1.25$ 16.25 She would have to make 17 shots.	



Teacher Note for Mari's Work

Students may wonder what finding the "smallest number" of shots to have a better percentage than Juan means.

Students can either identify that the proportion should have 14 shots instead of 13 shots OR that the final answer should be rounded up (as seen in the key).

12.2.4 Bring It Together: Proportions

Component Overview: Students summarize or synthesize their understanding of how to use proportions to solve problems in this activity.



12.2.4 Bring It Together: Proportions

1. Mistakes create opportunities to learn! Based on the mistakes you observed when examining other students' work, use the space provided to describe at least two things you learned and want to keep in mind when using proportions to solve problems.

**Means
I
Start
To
Acquire
Knowledge
Experience
Skills**

*Answers will vary.
Ensure the proportion is
set up correctly prior to
solving.
Ensure that your fraction
is correct when using the
scale factor to solve the
proportion.*



Utilize moments of reflection like this to cultivate growth mindset and self-regulation. Consider using sentence stems from growth mindset tools, MA.K12.MTR.1.1, or Universal Design for Learning Guidelines.

- Promote expectations and beliefs that optimize motivation (partner shout-outs).
- Facilitate personal coping skills and strategies (celebrate growth as much as or more than you celebrate proficiency).
- Develop self-assessment and reflection (recognize specific reflections that include examples).

12.2.5 Your Turn!

Component Overview: Students apply what they have learned to solve proportions in whatever way makes sense to them. Be sure to not force them to use one of the two methods presented. The key here is for their reasoning to be apparent in how they show their work.



12.2.5 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. Tyler swims at a constant speed of 5 meters (m) every 3 seconds. How long does it take him to swim 114 meters?

$$\frac{5 \text{ m}}{3 \text{ sec}} = \frac{114 \text{ m}}{x} \quad \boxed{x = 68.4 \text{ sec.}}$$

OR

$$114 \text{ m} \times \frac{3 \text{ sec.}}{5 \text{ m.}}$$

2. To make a shade of paint called jasper green, mix 4 quarts (qt) of green paint with $\frac{2}{3}$ cup (c) of black paint. How much green paint should be mixed with 4.5 cups of black paint to make jasper green?

$$\frac{\text{green}}{\text{black}} \left| \frac{4}{\frac{2}{3}} = \frac{?}{4.5} \right. \quad \boxed{27 \text{ quarts}}$$

3. It takes an ant farm 3 days to consume $\frac{1}{2}$ of an apple. At that rate, in how many days will the ant farm consume 6 apples?

$$\frac{3}{\frac{1}{2}} = \frac{x}{6} \quad \boxed{36 \text{ days}}$$



What similarities and differences do you see between the ways that you solved these problems?

Why did you decide to structure your proportions in these ways?

12.2.6 Wrap-Up: Reflection

Component Overview: Students complete a reflection about how well they think they can create and solve proportions.



12.2.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate where your current learning is in relation to today's learning target.

Answers will vary.

I can solve problems using proportions.

I need help.	I'm getting there, but	I understand this,
I can't get started.	I need more practice.	and I feel confident.



2. Explain why you placed your **X** where you did.

Answers will vary.



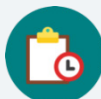
To help students break down where their strengths and weaknesses lie, consider including multiple scales with the following prompts.

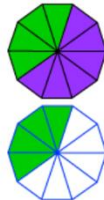
- I can set up a proportion correctly.
- I can solve a proportion correctly.
- I can determine if and when an answer should be rounded.

These results can also help you identify class trends for reteaching and remediation.

12.3 Solving Real-World Problems Involving Proportions

Lesson Introduction

 Benchmark	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.		 Learning Target	I can solve problems using proportions.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.		N/A	
 Essential Questions	- How can ratios and rates be used to relate and compare numbers in context? - What kinds of relationships can proportions represent? - How do you set up a proportion to represent a situation?			
 Lesson Narrative	As the third lesson in the section on solving problems with proportions, this lesson builds off of the prior two, introducing more complicated contexts and creating and solving part-to-part and part-to-whole comparisons with proportions. The lesson begins with an exploration of a part-to-part comparison using fractions as parts before moving to problems that have more context so that students can plan what is important to structure a proportion to solve.			
 Component Summary	12.3.1 Warm-Up (~5 min.)	Filling in the missing digits to create equivalent ratios (<i>SE p. 224</i>)	12.3.3 Bring It Together (10 min.)	Comparing and contrasting part-to-part and part-to-whole proportions (<i>SE p. 226</i>)
	12.3.2 Exploration (15-20 min.)	Exploring the approach of solving with proportions (<i>SE p. 225</i>)	12.3.4 Your Turn! (10 min.)	Practice creating part-to-part and part-to-whole proportions to solve problems (<i>SE p. 227</i>)
	12.3.5 Wrap-Up (~5 min.)	Proportion application in real-world context (<i>SE p. 228</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Calculators (Optional) Fraction/decimal conversion table	N/A

 Teacher Tips	Common Misconceptions - Students may struggle with differentiating between part-to-part vs part-to-whole comparisons in context. - Students may use direct division instead of proportional division. For example, if it takes 2 people 4 hours to do a certain task, students may mistakenly think that it would take 1 person 2 hours rather than 8 hours. - Students may use addition or subtraction instead of multiplication or division to find the scale factor.	Things to Consider - While still using proportional reasoning, we are no longer converting to different units. Making this clear for students who are looking for conversion rates means discussions are now focused on finding a missing piece of a ratio to complete the proportion so that the incomplete ratio has the same value of the given ratio. - In activities 12.3.2 Exploration, 12.3.4 Your Turn!, and 12.3.5 Wrap-Up, the scale factors are not whole numbers, which may hold some students up.
	Literacy and Language Routines Three Reads Students can practice Three Reads in 12.3.2 Exploration before applying it during 12.3.4 Your Turn!. Reading the problem and question multiple times with a specific purpose for reading supports students’ comprehension of the problem context, creates a plan for solving, and encourages meta-awareness of their use of math language.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure In 12.3.2 Exploration and 12.3.3 Bring It Together, students learn they need to read closely for relevant details of what each value in the problem means to uncover in the relationship between the values in order to create a proportion. The questions included help them see these relationships explicitly and flexibly in order to support both procedural and conceptual understanding.
	 Differentiate Instruction Consider visual scaffolding for comparing and contrasting part-to-part and part-to-whole relationships. Students will often be proficient in solving and calculating when given a proportion but struggle to find an entry point when interpreting a word problem. Consider using colors, household objects, anchor charts, and other strategies for creating as many entry points as possible for students to get started when problem-solving.	part-part or whole-part  6:4 part-part $\frac{10}{4}$ whole-part
	Lesson Notes:	

12.3.1 Warm-Up: Proportions with Fractions

Component Overview: Students use the structure of a proportion in a challenge to create a proportion using fractions as parts of the ratio. To limit the possibilities, each fraction is a unit fraction, and students can use each digit one to nine only once in the proportion.



12.3.1 Warm-Up: Proportions with Fractions

- Place a digit in each blue box to make the equality statement true. Use only digits 1 to 9, at most one time each.

$$\frac{\frac{1}{\boxed{2}}}{\frac{1}{\boxed{4}}} = \frac{\frac{1}{\boxed{3}}}{\frac{1}{\boxed{6}}}$$

Other possible answers include:

$$\frac{\frac{1}{\frac{2}{3}}}{\frac{1}{\frac{1}{6}}} = \frac{\frac{1}{\frac{4}{6}}}{\frac{1}{\frac{1}{9}}} \quad \frac{\frac{1}{\frac{6}{9}}}{\frac{1}{\frac{1}{3}}} = \frac{\frac{1}{\frac{2}{3}}}{\frac{1}{\frac{1}{6}}}$$



During this Warm-Up, have students describe their proportion answers verbally, using a structure like the following:

“ $\frac{1}{2}$ pizza is to $\frac{1}{4}$ pizza like $\frac{1}{3}$ pizza is to $\frac{1}{6}$ pizza.”

Work in more precise language like “equivalent” to replace words such as “like.”

12.3.2 Exploration: Which Information Should I Use?

Component Overview: In this Exploration, students are provided context where a part-to-whole proportion is necessary. Students first identify and describe the ratio given in the problem. Then students are told what the whole needs to be and create a new ratio that relates to the part-to-whole ratio $\frac{a}{40}$. Students then describe what the values in the proportion mean in relation to one another and solve. This procedure is repeated with another context in which three values make up the whole, including a question about the scale factor used for each part.



12.3.2 Exploration: Which Information Should I Use?

Elena is making large batches of fruit punch and marinated chicken for her upcoming family reunion.

She makes fruit punch by mixing 5 parts cranberry juice with 3 parts apple juice.

1. Complete the statement to explain what this means by writing numbers in each blank.

Regardless of the unit she uses to measure, Elena will use a ratio of cranberry to apple juice that is 5 to 3. For example, if she is measuring in cups, she would add 3 cups of apple juice for every 5 cups of cranberry juice she adds.

2. Elena plans to make 40 cups of fruit punch. Work with your partner to complete the proportion that could be used to find how much apple juice she needs for the fruit punch, a .

$$\frac{3}{8} = \frac{a}{40}$$

3. What does the numerator of each ratio represent in the proportion?

The amount of the apple juice

4. What does the denominator of each ratio represent in the proportion?

The total fruit punch

5. How did you use the values 5 and 3 to determine the numerator and denominator of the first ratio?

3 is apple juice only

Total points are $3 + 5 = 8$ parts



Three Reads

Consider a language routine like Three Reads to provide students scaffolding for making sense through the Exploration.

- *First Read:* Have students read through the Exploration in full without answering any questions.
- *Second Read:* Have students read through again, specifically circling or making note of questions that they are unsure how to unpack.
- *Turn and Talk:* Have students then compare with a partner or small group to clarify each other's questions and complete the activity.
- *Third Read:* Have students conduct a final read to reflect on what they have learned, ask questions, and correct any final errors.



Ask students what the 40 represents in the proportion and what it is being compared to. This positions students to see that this question is asking them to find the total cups of juice and not just three cups of apple juice or five cups of cranberry juice.

12.3.2 Exploration: Which Information Should I Use? (continued)

6. How much apple juice will Elena need to make 40 cups of fruit punch?

15 cups

The chicken marinade recipe she uses says, "Mix 3 parts oil with 2 parts soy sauce and 1 part orange juice." Elena needs to make 13.5 cups of marinade for the amount of chicken she's marinating.

7. Create a proportion that could be used to find the amount of oil Elena needs for the recipe.

$$\frac{\text{oil}}{\text{Total}} \bigg| \frac{3}{6} = \frac{?}{13.5 \text{ c}} \quad \boxed{6.75 \text{ cups}}$$

8. Elena is scaling the recipe up to make a larger batch. By what scale factor is Elena enlarging the recipe? Explain or show how you know.

$$\frac{13.5}{6} = 2.25 \text{ or } 2\frac{1}{4}$$

9. How could this scale factor be used to determine how many cups of each ingredient she needs to make the marinade?

Multiply each number of parts by 2.25

Oil: $3 \times 2.25 = 6.75$ cups oil

Soy Sauce: $2 \times 2.25 = 4.5$ cups soy sauce

Orange Juice: $1 \times 2.25 = 2.25$ cups OJ

10. Complete the summary by writing numbers on each blank based on the situation.
If she's measuring in cups, Elena can make 6 cups of marinade using the original recipe. She is scaling the recipe up by a scale factor of 2.25 to make 13.5 cups of marinade. To do this, she will mix 6.75 cups of oil with 4.5 cups of soy sauce and 2.25 cups of orange juice.



Consider bringing in measuring cups to provide students with visual cues or manipulatives for an additional entry point in making sense of setting up proportions.



Notice question 8 provides the connection between scaling up and multiplication. Consider prompting students to explain why scaling up means multiplication and what that means for proportional relationships.

- What is the relationship between the scale factor and the proportion?
- Would you always multiply using a scale factor in proportional relationships and problem-solving? Explain.

12.3.3 Bring It Together: Proportions

Component Overview: In this component, students encounter a formal introduction to the terms “part-to-part” and “part-to-whole.” They create a proportion for the context given for both a part-to-part question and a part-to-whole question and choose one to solve.



12.3.3 Bring It Together: Proportions

Recall that ratios can be used to describe part-to-part relationships or part-to-whole relationships.

Situation: Elena’s fruit punch recipe uses 5 parts cranberry juice for every 3 parts apple juice.	
Part-to-Part Example	Part-to-Whole Example
The ratio of cranberry juice to apple juice is 5:3.	The ratio of cranberry juice to total fruit punch is 5:8.

When setting up proportions to solve problems, it’s important to first determine which quantities need to be compared to find the missing value.

1. Complete the table.

Situation: A teacher is planning a field trip to the aquarium. The aquarium requires 2 chaperones for every 15 students. The teacher plans accordingly and orders a total of 85 tickets.	
Set up a proportion that can be used to <i>determine the number of students</i> the teacher is planning to have on the field trip.	$\frac{15}{17} = \frac{s}{85}$
Set up a proportion that can be used to determine the <i>number of chaperones</i> the teacher is planning to have on the field trip.	$\frac{2}{17} = \frac{c}{85}$

2. Solve only one of the proportions above.

c = 10 chaperone tickets

3. Explain how the solution to the proportion can be used to determine both the number of student tickets and the number of chaperone tickets the teacher ordered.
You can use the same scale factor, 5, to find the number of student tickets. OR Subtract 10 from 85 to find the number of student tickets.



In your own words, describe the difference between 5:3 and 3:5 in this context.



Give an example of a part: part relationship in this context. Give an example of part: whole relationship in this context.



First proportion: “S gives you the number of student tickets, which you can then subtract from the total of 85 to find the number of chaperone tickets.”

Second proportion: “The second proportion gives you a scale factor of 5, which you can use to multiply by the number of students (15) to find the total.”

12.3.4 Your Turn!

Component Overview: Students complete three problems that require them to identify the relationships between values to create a ratio and use that ratio to create and solve a proportion in both part-to-part and part-to-whole contexts.



12.3.4 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar. How much oil is needed to make 19 teaspoons (tsp) of salad dressing?

$$\frac{3}{4} = \frac{x}{19} \quad \boxed{14 \frac{1}{4} \text{ tsp.}}$$

2. A grocer can buy strawberries from a local farmer for \$1.38 per pound (lb). His last strawberry order cost \$241.50. How many pounds did the grocer order?

$$\$241.50 \times \frac{1 \text{ lb.}}{\$1.38} = \boxed{175 \text{ lb.}}$$

3. Andre and Han are moving boxes. Andre can move 4 boxes every half hour. Han can move 5 boxes every half hour. How long will it take Andre and Han to move all 72 boxes?

$$\frac{30 \text{ min}}{9 \text{ boxes}} = \frac{x}{72 \text{ boxes}} \quad \boxed{\begin{array}{c} 240 \text{ minutes} \\ \text{or} \\ 4 \text{ hours} \end{array}}$$



Consider extending the Three Reads language routine to these questions for students who still need support. interpreting word problems and constructing proportions.

- *First Read:* Focus on what the question is asking.
- *Second Read:* Focus on units and values included.
- *Third Read:* Focus on representing the values and question using a proportion.

12.3.5 Wrap-Up: Ticket Out the Door

Component Overview: This Wrap-Up assesses students' abilities to create and solve a proportion using given information resulting in part-to-part comparisons and part-to-whole comparisons within the same problem context.



12.3.5 Wrap-Up: Ticket Out the Door

A recipe for sparkling grape juice calls for $1\frac{1}{2}$ quarts of sparkling water and $\frac{3}{4}$ quart of grape juice.

1. How much sparkling water should be mixed with 9 quarts of grape juice?

$$\frac{1.5}{0.75} = \frac{?}{9} \quad \boxed{18 \text{ quarts}}$$

2. How much grape juice should be mixed with $\frac{15}{4}$ quarts of sparkling water?

$$\frac{1.5}{0.75} = \frac{3.75}{?} \quad \frac{\frac{3}{2} \times \frac{5}{2}}{\frac{3}{4} \times \frac{5}{2}} = \frac{\frac{15}{4}}{?} = \frac{15}{8}$$

$$\boxed{1.875 \text{ qt. or } 1\frac{7}{8} \text{ qt.}}$$



Students should continue to build fluency in operations with fractions. However, when evaluating student work, consider giving two scores, one for setting up proportions and one for calculations.

This differentiated approach will not only give you more specific information per student, but will also give students more specific feedback on their work.

12.4 Discounts and Markups

Lesson Introduction

 Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.		 Learning Target	I can solve problems involving discounts and markups.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.		N/A	
 Essential Questions	- What are the similarities and differences between calculating discounts and markups? - How do you solve real-world problems involving discounts and markups?			
 Lesson Narrative	In this lesson, students solve problems involving discounts and markups. The lesson begins by exploring discounts and markups with familiar contexts to access students' prior knowledge. Students then work with a partner to solve problems they select from a menu of options. The lesson concludes with a student reflection that can be used to plan for upcoming lessons, as students continue working with percentages.			
 Component Summary	12.4.1 Warm-Up (5-8 min.)	Reflect on how civil engineers use math (SE p. 229)	12.4.3 Activity (15-20 min.)	Collaboratively select and solve real-world discount and markup problems (SE p. 232)
	12.4.2 Guided Instruction (15 min.)	Explore how to determine discounts and markups in context (SE p. 229)	12.4.4 Wrap-Up (~5 min.)	Reflect on which parts of the lesson were difficult or confusing (SE p. 234)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Discount <u>Additional Terms:</u> - Markup		Career Profile video	- Consider bringing in receipts and sales flyers to show students the utility of the mathematics in the real-world. 12.4.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may confuse the discount or markup amount, original cost, and final cost of a product when determining how to solve a problem.	- Facilitate a conversation with students about the differences between the discount/markup amount, original cost, and the final cost of a product. Model how to approach problems if needed and encourage students to mark up the text as they read and make sense of each problem. - During the 12.4.2 Guided Instruction, consider creating and displaying an anchor chart with instructions, examples, and visuals for students to reference that highlights different approaches to solving problems.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 12.4.3 Activity, ask students to take turns selecting menu items before they work together to solve the problems and write their work in the tables. The student who selects the problem should begin the discussion about how to approach the problem. Discussing the problems with partners will build student confidence and understanding. It also holds both students accountable.	MA.K12.MTR.5.1: Patterns and Structure In 12.4.3 Activity, students solve problems with discounts and markups. Students can use a variety of strategies. Ask students to use their reasoning and sense-making skills to determine what the problem is asking them to find and what information is provided and develop a plan to solve. Prompt students to look for patterns and structure among the discount and markup problems they solve.
 Differentiate Instruction	<i>For 12.4.2 Guided Instruction, consider providing visual and kinesthetic entry points by having students look at sales flyers to see real-world examples of discounts.</i> <i>Discussions during 12.4.3 Activity provide auditory entry points. Consider previewing the problems and having an eye on which students are selecting from the menu so that you can help push students' thinking as appropriate.</i>	
Lesson Notes:		

12.4.1 Warm-Up: Civil Engineer

Component Overview: Play the Career Profile video on the civil engineer for the entire class. After students watch the video (2:49 minutes), prompt them to reflect on the profession by answering the two questions independently.



12.4.1 Warm-Up: Civil Engineer

Alishia Ballard works as a civil engineer for San Francisco Public Works. Civil engineers work on the plans to create structures such as bridges, large buildings, etc. They also analyze whether or not structures are safe.

1. How do you think civil engineers use math in their tasks?

Answers will vary. Civil Engineers use math to calculate how much material will be needed to create a structure. Civil engineers use math to figure out what dimensions are needed to make sure the structure does not collapse.

Alishia enjoys working at Public Works because it's a way to give back to her community. Every structure plan she works on or structure she inspects is all for the people around her.

2. What is something you have done that gives back to your community, your school, or your peers?

Answers will vary.



Do you know someone who is a civil engineer or works in public works? Consider having a guest speaker join your class (virtually or in person). If time permits, consider having students return to this lesson for a project on the careers explored throughout the year.

12.4.2 Guided Instruction: Discounts and Markup

Component Overview: Students are guided through a series of questions, fill-in-the-blank statements, and problems in real-world contexts involving discounts and markups. The questions and problems provide students with strategies that can be used to determine discounts and markups by accessing students' prior experiences and without using a formula.



12.4.2 Guided Instruction: Discounts and Markup

Percentages and ratios are often used to express or solve problems involving changes in the price of goods and services.

1. The price of a shirt is \$18.50. Natalie has a coupon that reads "discount $\frac{1}{2}$ off."
 - a. With the coupon, will Natalie pay more or pay less than \$18.50 for the shirt?
Less because a coupon lowers the price.
 - b. Discuss with your partner what " $\frac{1}{2}$ off" means. Summarize your discussion.
Half off means to take away half of the price.

A **discount** is a reduction of a price for goods or services.

- c. By what percentage will the coupon reduce the price?
50%
- d. What is the discount amount, in dollars, taken off the shirt price?
\$9.25
- e. What is the discounted price, in dollars, of the shirt after the coupon is applied?
\$9.25



Students are introduced to the concept of discounts. Consider asking students to write the definition and examples in their notes or in an interactive journal. Also, facilitate a discussion about situations where discounts and markups are used in the real-world. Help students identify keywords or phrases that they can use to determine whether an item is being discounted or marked up.



- What percentage is equivalent to $\frac{1}{2}$ off?
- How can converting between equivalent forms of rational numbers help you solve discount and markup problems?

12.4.2 Guided Instruction: Discounts and Markup (continued)

2. Cassidy, a young inventor of Baby Toon, appeared on Shark Tank in 2019. Cassidy shared that it costs her \$6.60 to produce one of her baby spoons and that she sells them to customers marked up to \$15.00.

- a. Does a markup increase or decrease the original price?

Increase

- b. Discuss with your partner what you think markup means. Summarize the discussion by writing a definition for markup.

Answers will vary. The definition of markup is the increase of a number by a certain percentage. A markup is when a new number is created by adding a percentage of the original number to the original number.

- c. Complete the statements.

A markup is applied to goods or services so that the

seller is able to make a ☐ loss. ☐ profit. The markup

- ☐ increases
☐ decreases

the price from the cost to the seller and

is usually determined using percentages.

- d. If Cassidy placed a 100% markup on her spoons that only cost \$6.60 to produce, what would the price of a spoon be for customers?

\$13.20

- e. Discuss with your partner how you determined a 100% markup of \$6.60. After your discussion, make any revisions to part b, if necessary.

100% of \$6.60 is \$6.60. When the percentage amount is added to the original number, the sum is \$13.20.



Consider providing supports and different entry points for students struggling with the concepts of discounts and markups. For example:

- Dollar and coin manipulative tools can be used to provide visual and kinesthetic entry points for problems involving money (question 2).
- Provide an auditory entry point by allowing students to work with a partner or by modeling through a Think Aloud.

12.4.2 Guided Instruction: Discounts and Markup (continued)

- f. Jovi and Malik used two different methods to find the price of the spoon if the markup used is 61%. Part of their work is shown.

Jovi's Work	Malik's Work
$\frac{m}{6.60} = \frac{61}{100}$ $100 \cdot \frac{1}{6.60} \cdot \frac{m}{6.60} = \frac{61}{100} \cdot \frac{1}{100} \cdot 100 \cdot 6.60$ $100m = 402.6$ $m = 4.026$	$0.61(6.60) = m$ $4.026 = m$

Complete the statements.

Jovi used a percent ☐ equation ☒ proportion to find the value

of m while Malik used a percent ☐ equation. ☒ proportion. Both

methods resulted in the same value of m , which

represents the ☐ dollar amount of the markup. ☒ price of the spoon after markup.

At a 61% markup, the spoon would sell for ☐ \$4.02. ☐ \$4.03. ☐ \$10.62. ☒ \$10.63.

- g. Jovi and Malik both found the price of the spoon after a 61% markup by calculating the amount of the markup and adding it to the cost of the spoon. What percentage could they have used if they wanted to calculate the price after markup in one step?
161%



Consider asking students the following:

- What is the same and different about Jovi's and Malik's work?
- What does the value they calculated tell you about the markup and final cost of the spoon?



Challenge students to write a numeric expression representing the cost of the spoon after a 61% markup.

$6.6 + 0.61(6.6)$ or $1.61(6.6)$

12.4.3 Activity: Math Menu

Component Overview: Students work with a partner to solve real-world discount and markup problems. Students are allowed to choose their preferred problems from the categories in a math menu.



12.4.3 Activity: Math Menu

With your partner, choose your meal from the math menu shown. Select one appetizer, one main course, two side dishes, and one dessert. Put a * next to your selections. Then, solve each problem selected.

Appetizers
A1. A book has been discounted by $\frac{3}{10}$. The original price of the book is \$34.99. What is the amount being discounted from the book?
A2. Westside Skate Shop is having a storewide $\frac{1}{5}$ off sale. Find the amount saved on a \$66 skateboard.
A3. Jennea bought a scratched iPhone because it was discounted $\frac{9}{20}$ off the original price of \$795. What was the reduced price, in dollars?
Main Courses
M1. Ramin's Discount Automobiles buys a used car for \$3,000. They mark it up 65% before selling it to a customer. What was the sale price of the car?
M2. Elliott sells grab-and-go lunches. He spends \$2.70 per lunch on ingredients and then adds a 110% markup to sell. What is the selling price of one sandwich?
M3. It costs JS Surfboards \$225.00 in labor to make one Model S surfboard. The company marks up the surfboard 51% to sell. What amount, in dollars, are Model S surfboards sold for?



Take Turns

Ask students to take turns selecting menu items before working together to solve the problems and writing the work in the tables. The student who selects the problem should begin the discussion of how to approach it. Discussing the problems with partners will build student confidence and understanding. It also holds both students accountable.

12.4.3 Activity: Math Menu (continued)

Side Dishes

- S1. A concert ticket is on sale for \$129. The sign says 14% off the original price. What was the original price of the ticket?
- S2. A scientific calculator is 15% off. The sale price of the calculator is \$13.99. What is the original cost of the calculator?
- S3. The price of an apple is \$1.25. If you get a 10% discount, how much do you have to pay?
- S4. Teena bought a bicycle for \$528 after getting a discount of 12% taken off its normal price. Find the normal price of the bicycle.

Desserts

- D1. A store sells notebooks for \$1.60 after a $\frac{3}{10}$ markup. How much is the markup?
- D2. Burgermonster sells its burgers for \$1.84. To determine the selling price, the restaurant added a markup of $\frac{23}{10}$ to the cost of the ingredients used to make a single burger. What is the cost Burgermonster pays for ingredients?



- How can you determine whether the problem involves a discount or markup?
- What are the similarities and differences between the four choices for side dishes?
- Which problem would you prefer to solve? Why?



Some students may struggle completing the number of questions in this activity during one class session. Consider decreasing the number of side dishes to one question if needed.

12.4.3 Activity: Math Menu (continued)

Appetizer Selection Work and Answer

A1: $(34.99) \frac{3}{10} = \$10.50$ is being discounted from the book

A2: $\frac{1}{5}(66) = \$13.20$ is being saved on the skateboard

A3: $\frac{11}{20}(795) = \$437.25$ was the reduced price

Main Course Selection Work and Answer

M1: $3000(1.65) = \$4,950.00$ is the sale price of the car

M2: $2.70 + (1.10)(2.70) = \$5.67$ is the selling price of a sandwich

M3: $225(1.51) = \$339.75$ is the sale price of surfboards

Side Dish Selection Work and Answers

S1: $\frac{86}{100} = \frac{129}{x}$; $x = \$150.00$ was the original price of the ticket

S2: $\frac{85}{100} = \frac{13.99}{x}$; $x = \$16.46$ was the original cost of the calculator

S3: $\frac{9}{10}(1.25) = \$1.13$ is the price you pay for the apple

S4: $\frac{88}{100} = \frac{528}{x}$; $x = \$600.00$ is the normal price of the bicycle

Dessert Selection Work and Answers

D1: $\frac{1.3}{1} = \frac{1.6}{x}$; $x \approx \$1.23$
 $1.60 - 1.23 = \$0.37$ is the markup

D2: $1.84 = 3.3x$; $x = \$0.56$ is the cost of the ingredients



Students can solve the problems in different ways using what they have learned about percentages. The sample work shown in the key is not the only strategy.



Ask students that finish early to create their own discount or markup problem based on a real-world situation. Bring in receipts and sales flyers that can serve as an inspiration for students' problems.

12.4.4 Wrap-Up: Reflection

Component Overview: Students reflect on what may have been difficult or confusing as they worked with discount and markup problems. Consider student responses when planning for the next lesson.



12.4.4 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on discounts and markup?



Answers will vary.



For students who struggle expressing themselves in written form, consider alternative methods, such as:

- Using a rating system (i.e. 1=no confusion, 2=some confusion, 3=a lot of confusion) to label the questions from the lesson
- Writing a list of confusing topics
- Circling questions they found difficult in their book

12.5 Determining Tax, Tip, or Fees Given a Percentage

Lesson Introduction

 Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.		 Learning Targets	- I can determine the amount of tax, tip, or fee given a percentage. - I can determine the percentage of a discount or markup.
 Benchmark Coherence	Building On			Working Toward
	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.			N/A
 Essential Questions	- How do you determine the percentage of a discount or markup? - How do you calculate tax, tips, or fees for a given percentage?			
 Lesson Narrative	In this lesson, students extend their knowledge and skills with finding discounts and markups to determine the percentage used to discount or markup. Students also explore how to calculate taxes, tips, and fees in context.			
 Component Summary	12.5.1 Warm-Up (~5 min.)	Error analysis of sales flyer (<i>SE p. 235</i>)	12.5.4 Guided Instruction (10-15 min.)	Explore how to find taxes, tips, and fees (<i>SE p. 237</i>)
	12.5.2 Guided Instruction (10 min.)	Explore methods to determine the percentage of a discount or markup (<i>SE p. 235</i>)	12.5.5 Guided Practice (5 min.)	Practice determining tips, taxes, and fees (<i>SE p. 239</i>)
	12.5.3 Try It (5 min.)	Error analysis of calculating a percentage discount (<i>SE p. 237</i>)	12.5.6 Your Turn! (5 min.)	Practice finding taxes and percentages of discount and markup (<i>SE p. 240</i>)
	12.5.7 Wrap-Up (~5 min.)	Demonstrate ability to calculate the price of an order after tax (<i>SE p. 240</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Gratuity (Tip) - Markup - Discount - Markdown - Commission - Sales Tax - Subtotal - Total		<u>Additional Preparation</u> - Consider which problems or components you may want students to solve without using a calculator. - Consider what calculations/work you want students to show in their workbook for problems where they use a calculator.	

 Teacher Tips	Common Misconceptions • Students may struggle converting between different forms of rational numbers (i.e., fraction to decimal, percent to decimal, decimal to percent). • Students may not understand the difference between the subtotal and the total and which is used when determining tax or tip amounts in various contexts.	Things to Consider • During the 12.5.2 Guided Instruction, consider creating an anchor chart with instructions, examples, and visuals for finding percentage of a discount and markup. • Allowing students to use calculators during the lesson will help students focus on discovery and process of calculating discounts, markups, taxes, tips, and percentages. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers. • There are multiple strategies for solving the problems in this lesson. Encourage students to use strategies that make sense for them as they build fluency in solving problems.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Algebra Talk In 12.5.1 Warm-Up, considering having students initially find and correct the error in the problem independently. Next, engage students in a discussion about the strategies students used to correct the error. This routine can also be used in debriefing 12.5.3 Try It.	MA.K12.MTR.6.1: Reasonableness In 12.5.6 Your Turn!, students determine the percentage of a discount and markup and find the tax in a real-world situation. Prompt students to informally check their answers by considering if the solution makes sense given the context. Then have students also formally check their calculations for accuracy.
	In 12.5.2 and 12.5.4 Guided Instructions, components prompt students to annotate the examples using the Mark-Up strategy. Consider modeling this strategy at the conclusion of 12.5.4 Guided Instruction.	
 Differentiate Instruction		
Lesson Notes:		

12.5.1 Warm-Up: Math Fail

Component Overview: Students identify and correct error(s) on a sale sticker. If students do not initially recognize the error(s), prompt them to calculate the discount and compare their work and answer to the discount on the sticker.



12.5.1 Warm-Up: Math Fail

The following sticker was placed on an item for sale.



1. Describe any error(s) on the sticker.
Answers will vary. Half price of \$129 is not \$99, it is \$64.50. The sale price should be \$64.50.
2. Correct the sticker so that it accurately reflects the "1/2 Price" advertised in the first line.
\$64.50



Algebra Talk

If time allows, consider using this language routine before moving on to the next component.

After students have completed the Warm-Up independently, ask students to engage in a discussion about the strategies they used to find and correct the error.

12.5.2 Guided Instruction: Finding the Percentage of a Discount or Markup

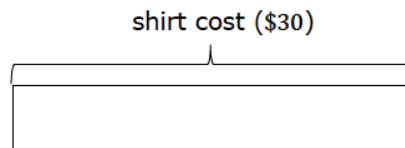
Component Overview: Students are guided through a series of questions as they develop a stronger understanding of how to find discounts and markups in real-world scenarios. Students find the discount and markup amounts, final amounts, and the percentages of the discounts and markups.



12.5.2 Guided Instruction: Finding the Percentage of a Discount or Markup

Recall that discount and markup problems typically include given percentages.

- The tape diagram shows the original cost of a shirt, \$30.

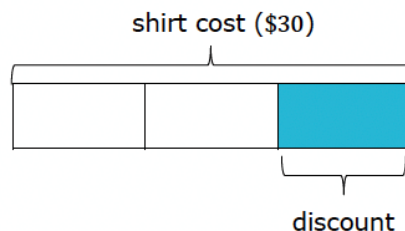


A store manager wants to sell more of this shirt before the end of summer. The manager decides to put the shirt on sale for $\frac{1}{3}$ off the original price.

- Will customers pay more or less than the original price for the shirt?

Less

The tape diagram is divided into thirds.



- What is the price of the shirt after the discount?
- How much will customers save from the original price?
- Discuss with your partner how to calculate the percentage discounted if given the original price of the shirt and the sale price of the shirt.

$$\frac{x}{100} = \frac{10}{30}; x = 33\%$$

I used a percent proportion with the discounted amount to find what percentage 10 is of 30.



Consider creating an anchor chart with instructions, examples, and visuals for finding the percentage of a discount and markup.

Also, prompt students to underline keywords and annotate the examples in their own words. Debrief each component with a whole-class discussion to provide an auditory entry point.



- How does the tape diagram show $\frac{1}{3}$ of the original price of the shirt?
- How do you know whether the customer will pay more or less than the original price from the question 1?



Students could also use an equation to determine the percentage discounted.

$30x = 10$, where x is the percentage discounted represented as a decimal.

12.5.2 Guided Instruction: Finding the Percentage of a Discount or Markup (continued)

- e. If the \$30 shirt was discounted to \$18, what percentage of the shirt was discounted.

40%

2. Nova buys earrings from a supplier for \$13.20. She sells them in her boutique for \$24.95. Clare works at the boutique and was curious what percentage markup Nova uses. The table shows 2 different methods Clare used to determine the percentage the earrings are marked up when sold.

Method 1	Method 2
$\$24.95 = (1 + \text{Markup rate}) \times \13.20 $\frac{24.95}{13.20} = 1 + \text{Markup rate}$ $1.89 = 1 + \text{Markup rate}$ $1.89 - 1 = \text{Markup rate}$	Nova paid \$13.20 Nova is selling for \$24.95 $\$24.95 - \$13.20 = \$11.75$ What % is 11.75 of 13.2? $\frac{11.75}{13.2} = 0.89$

- a. Compare both methods. Discuss the similarities and differences in the two methods with your partner. Then, write them in the table.

Similarities	Differences
<i>In both methods, Clare is trying to determine the markup rate. She finds the same markup rate using both strategies.</i>	<i>Method 1 considers how the store computed the new sales price whereas method 2 finds the mark-up and determines what proportion of the original price this represents.</i>

- b. What is the percentage of the markup?
89%



Consider giving students the opportunity to reflect on whether they prefer one strategy to the other based on the methods shared by Clare.

12.5.3 Try It: Finding the Percentage of a Discount or Markup

Component Overview: Students identify and correct error(s) in Johnetta's calculations of the discount percentage of a backpack.



12.5.3 Try It: Finding the Percentage of a Discount or Markup

1. Johnetta calculated the percentage discounted for a backpack that normally costs \$25 but is on sale for \$21. Johnetta has made errors in her work. Identify Johnetta's errors and correct her work.



If students do not initially recognize the error(s), prompt them to calculate the discount percentage and compare their work and answers to Johnetta's.

Backpack original cost is \$25
 Backpack sale cost is \$21

$$\frac{\$25}{\$21} = 1.19$$

 Discount 1.19%

Answers will vary. Johnetta should have found the amount of the discount, \$4, by subtracting 21 from 25. Then, she should have divided the difference by 25 and multiplied by 100 to get 16%.

$$\frac{\$4}{\$25} = 0.16$$

Discount 16%

She could have also set up a percent proportion or equation to solve.

$$\frac{(25 - 21)}{25} = \frac{x}{100}$$

$25x = 25 - 21$, where x is the percentage discount represented as a decimal



Algebra Talk

Have students initially find and correct the error in the question independently. Next, engage students in a discussion about the strategies they used to correct the error.

12.5.4 Guided Instruction: Taxes, Tips, and Fees

Component Overview: Students are guided through a series of questions as they develop an understanding of how to calculate taxes, tips, and fees in real-world scenarios. Explain the difference between the subtotal and the total to avoid student confusion.



12.5.4 Guided Instruction: Taxes, Tips, and Fees

1. Baxter has been saving all of his money to purchase new Nike Air Max sneakers. After four months, he finally had \$200 saved, so he went to purchase the sneakers. Unfortunately, when Baxter checked out, the total came to \$212, which was \$12 more than he had saved.

Subtotal	\$200.00
Estimated Shipping	\$0.00
Estimated Tax	\$12.00
TOTAL	\$212.00

- a. Discuss with your partner why Baxter's total was more than \$200.
The total is more than \$200 because he had to pay sales tax.
- b. If Baxter had to pay 8% sales tax on the same Nike Air Max sneakers, what would the cost of the taxes be?
 $200 \times 0.08 = \$16$
- c. Explain whether the estimated tax from Baxter's online search was more or less than 8%.
The online sales tax was less because it was less than \$16. It was only 6% of the total.



Facilitate a brief discussion about how and when they have used or heard of taxes, tips, and fees in their everyday lives.

12.5.4 Guided Instruction: Taxes, Tips, and Fees (continued)

Work with your partner to complete the following.

2. The Cardone Family is planning to have a birthday celebration at one of their favorite restaurants. The restaurants they are considering are in different counties that each have different sales tax rates.
 - a. Complete the table to determine how much tax will be at each restaurant given the same pre-tax cost of dinner. The first row of the table has been completed.

Restaurant	Cost of Dinner (Pre-tax)	How Much Sales Tax?
Restaurant A	\$120	Sales Tax 7% $(120)(0.07) = 8.40$ \$8.40 tax
Restaurant B	\$120	Sales Tax 7.5% \$9
Restaurant C	\$120	Sales Tax 8% \$9.60
Restaurant D	\$120	Sales Tax 8.5% \$10.20

- b. The Cardone Family plans to leave a tip for their server. They intend to tip 20% of the cost of their dinner. If the cost of dinner is \$120, how much tip will the server receive?
\$24
3. Qualyen bought a new computer and printer online totaling \$850. He had to pay an additional 3% in shipping fees on top of the \$850. With your partner, determine the cost of the shipping fees.
\$25.50



- What do you notice about the options for the Cardone family dinner?
- What are the similarities and differences in the restaurant choices?
- Based on the information provided, which restaurant is in the county with the largest tax rate? The smallest?



Challenge students to create equations that can be used to calculate the tax, tip, and fee. Prompt students to check their equations and make any necessary revisions.

12.5.4 Guided Instruction: Taxes, Tips, and Fees (continued)

There are many situations where a percentage of an amount of money is added to or subtracted from a total amount. For example:

- If a restaurant bill is \$34 and the customer pays \$40, they left \$6 dollars as a tip for the server. That is 18% of \$34, so they left an 18% tip.
- If the price tag on a shirt in Arizona is \$11.50, then the sales tax is $(0.075) \cdot 11.5 = 0.8625$, which rounds to 86 cents. The customer pays $11.50 + 0.86$, or \$12.36 for the shirt.
- A student goes to a hairdresser to get ready for the 8th grade formal dance. The total is \$125. The student's parents tip the hairdresser 15%. The total cost of the service is \$143.75.
- A gentleman purchases a suit originally priced at \$397.98. The suit is on clearance with a 45% off sticker. The discount reduces the suit \$179.09 off the original price, resulting in the new suit price being \$218.89.



Encourage students to use the Mark-Up strategy as they read through the sample situations provided in this component.

Consider also structuring this with a Think Aloud for struggling readers where the teacher models how to mark up the text with annotations for future reference.

	Goes to	How It Works
Sales Tax	The government	Added to the price of the item
Gratuity (tip)	The person providing the service	Added to the cost of the service
Markup	The seller	Added to the cost of an item so the seller can make a profit
Markdown (discount)	The customer	Subtracted from the price of an item to encourage the customer to buy it
Commission	The salesperson	Percentage of the value of the sale that is subtracted from the total sale and given to the salesperson

12.5.5 Guided Practice: Calculating Tax, Tip, and Fees

Component Overview: Students practice determining tip, tax, and fees in real-world scenarios.



12.5.5 Guided Practice: Calculating Tax, Tip, and Fees

1. Johnny went shopping for his mom at Save More Foods and his subtotal was \$11.75. The tax applied at the grocery store is 6%. How much was the tax for his purchase?
\$0.71
2. Rosa and her friends are eating at a restaurant for dinner. Their bill was \$45.80. They want to leave a 20% tip. How much tip will the server receive?
\$9.16
3. Rick bought 3 shirts for \$18 each, 2 pairs of socks for \$3.99 a pair, and a pair of slacks for \$45.00. The sales tax rate is 8.5%. How much sales tax did Rick pay?
\$9.09



How do you rewrite 6% as a decimal?

Use mental math to determine 10% of \$45.80. How can you use this value to determine the tip the server will receive in question 2?

12.5.6 Your Turn!

Component Overview: Students are provided with additional practice calculating sales tax and the discount and markup percentages.



12.5.6 Your Turn!

1. A city in Ohio has a sales tax rate of 7.25%. Complete the table to determine how much sales tax is for each item.

Item	Price Before Tax (\$)	Sales Tax (\$)
Pillow	8.69	\$0.63
Blanket	22.00	\$1.60
Trash Can	14.50	\$1.05

2. A store has a sale on pants. The original price of the pants is \$19.00. With the discount, the price of one pair of pants before tax is \$15.20. What was the percent discount?

20%

3. A music store marks up its instruments. The store bought a guitar for \$45 and sold it for \$58.50. The store bought a trumpet for \$80 and sold it for \$104. What is the percent markup the store used to price the guitar and trumpet?

30%



Consider asking students the following:

- How did you calculate the sales tax?
- Why did you select this method?
- Will this method always work?

12.5.7 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their ability to calculate the amount of sales tax applied to an order. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should be allowed to access a calculator to complete the Wrap-Up.



12.5.7 Wrap-Up: Ticket Out the Door

- Mrs. Jones attended a Southern Living home party. She purchased a vase that cost \$34.99 plus \$3.00 for shipping. There is a 8.25% sales tax on the cost of the vase and shipping charge.

What is the amount of tax Mrs. Jones paid on her purchase?

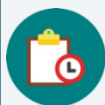
\$3.13



Remind students that they can reference similar problems they have solved in the lesson. Prompt students to check the reasonableness of their answers.

12.6 Solving Multi-Step Problems Involving Discounts, Taxes, Tips, or Fees

Lesson Introduction

 Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.		 Learning Target	I can solve multi-step problems that involve discounts, markups, taxes, tips, or fees.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.		N/A	
 Essential Questions	- How do you solve multi-step problems that involve discounts, markups, taxes, tips, or fees? - How can you determine the amount of sales tax or percent gratuity given a bill subtotal and final total?			
 Lesson Narrative	In this lesson, students solve multi-step real-world problems involving discounts, markups, taxes, tips, or fees. This lesson builds on students' prior work solving one-step discount, markup, tax, tip, and fee problems.			
 Component Summary	12.6.1 Warm-Up (~5 min.)	Consider percentages on a double number-line model (<i>SE p. 241</i>)	12.6.3 Collaboration (10-12 min.)	Collaboratively solve problems involving taxes, tips, and totals (<i>SE p. 243</i>)
	12.6.2 Exploration (10-15 min.)	Explore finding sales tax rates and tipping in real-world contexts (<i>SE p. 242</i>)	12.6.4 Your Turn! (10-13 min.)	Practice solving problems involving taxes, tips, fees, discounts, and markups (<i>SE p. 245</i>)
	12.6.5 Wrap-Up (~5 min.)	Demonstrate understanding of solving multi-step sales tax, discount, markup, and fee problems (<i>SE p. 246</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Gratuity (Tip) - Markup - Discount - Sales Tax - Subtotal - Total		- Calculators (Optional) Chart paper (Optional) Markers	- Consider which problems or components you may want students to solve without using a calculator. - Consider what calculations/work you want students to show in their workbook for problems where they use a calculator.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle converting between different forms of rational numbers (i.e., fraction to decimal, percent to decimal, decimal to percent). - Students may not understand the difference between the subtotal and the total and which are both used when determining tax or tip amounts in various contexts.	- During the 12.6.2 Exploration, consider modeling a portion of problem if needed, providing specific instructions on strategies that can be used to find a sales tax percentage. Listen as students collaborate on question 1a to determine if this is necessary. - During the 12.6.4 Your Turn!, facilitate a discussion with students explaining that tips can be based on the subtotal (without tax) or based off the total (with taxes). - Allowing students to use calculators during the lesson will help students focus on discovery and process of calculating discounts, markups, taxes, tips, and percentages. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers. - There are multiple strategies for solving the problems in this lesson. Encourage students to use strategies that make sense for them as they build fluency in solving problems.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think Pair Share In 12.6.3 Collaboration, consider asking students to initially complete the problems independently. Then have students discuss their work with a partner. These discussions will help students gain confidence and understanding.	MA.K12.MTR.6.1: Reasonableness In 12.6.4 Your Turn!, students solve multi-step problems using a variety of strategies. Prompt students to informally check their answers by considering if the solution makes sense given the context. Then have students also formally check their calculations for accuracy.
 Differentiate Instruction	Consider building upon any anchor charts created in earlier lessons to add additional content learned. Consider modifying the 12.6.3 Collaboration and 12.6.4 Your Turn! into stations that students can work through with a partner or small group. Determine how to set up the rotation and which stations may require more support from the teacher as students work through them.	
Lesson Notes:		

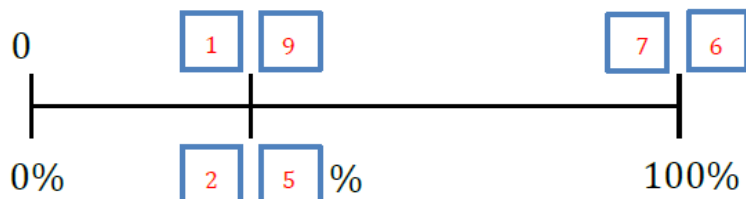
12.6.1 Warm-Up: Percentages on a Linear Model

Component Overview: Students create an accurate number line using percentages of a whole by filling the digits one through nine into the boxes marked on the number line.



12.6.1 Warm-Up: Percentages on a Linear Model

1. Fill in the blue boxes using the digits 1 – 9, one time each at most, to create an accurate number line.



Students have worked with double number-line models before.

If time allows, consider having students discuss question 2b and/or share strategies/solutions for question 1.

2. There are multiple possible solutions.
 - a. Find at least one other solution.
17, 68, 25%
 - b. Explain whether using 47% for the percentage makes sense.
No, because the line would need to be closer to the middle since 47 is close to 50 which is halfway.

12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts

Component Overview: Students explore how to find sales tax rates and how to tip in a real-world context.



12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts

1. Cities often have different sales tax rates based on local government decisions.

- a. Discuss with your partner how you might find the percentage of sales tax if you are given the price of an item and the amount of sales tax paid for the item. Summarize your discussion.

If you divide the sales tax by the price, you can find the rate. You could also set up a percent proportion to find the rate.

- b. With your partner, complete the two tables to determine the sales tax rates for City A and City B.

City A			
Item	Price (\$)	Sales Tax (\$)	Sales Tax Rate (%)
Paper Towels	8.99	0.63	7%
Laundry Soap	11.83	0.83	7%

City B			
Item	Price (\$)	Sales Tax (\$)	Sales Tax Rate (%)
Paper Towels	8.99	0.54	6%
Laundry Soap	11.83	0.71	6%

- c. Determine the total cost, with tax, of paper towels and laundry soap in City A.

\$22.28

- d. The pre-tax cost of 5 beach towels in City A is \$54.95. Determine the total cost, with tax, of 5 beach towels in City A.

\$58.80



If needed, consider doing a Think Aloud as you solve question 1b (only City A) after giving students an opportunity to discuss question 1a. This will provide a model for students as they try the other problems.



Is the sales tax rate for both items in one city the same or different? Why would this make sense in context?

12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts (continued)

- e. If the subtotal for the purchase of a new bathing suit in City B is \$47.97, what is the total cost of the bathing suit with tax?

\$50.85

- f. Find the total cost, with tax, on paper towels, laundry soap, and a new bathing suit in City B.

\$72.92

2. Jada and Corrin had lunch at a local restaurant. The final bill for their meal is shown.

They decide to tip their server 18% of the subtotal. Jada says the tip should be \$7.56, but Corrin disagrees. Corrin calculated the tip to be \$8.28.

- a. Explain who calculated the correct tip.

Jada correctly calculated the tip because \$42.00 times 0.18 is \$7.56. Corrin calculated the tip from the total which includes taxes, instead of from the subtotal.

Date: Sep. 12th
Time: 6:55 PM
Server: # 27

Bread Stix	9.50
Chicken Parm	15.50
Chef Salad	12.00
Lemon Soda	2.00
Tea	3.00

Subtotal	42.00
Sales Tax	3.99
Total	45.99

- b. What is the total amount, including the tip, that the friends paid at the restaurant?

\$53.55



Prompt students to calculate the tip and then compare their answers to Jada's and Corrin's answers. Remind students to convert the tip percentage to a decimal.

12.6.3 Collaboration: Taxes, Tips, and Totals

Component Overview: Students collaboratively determine taxes, tips, markups, and discounts in real-world contexts.



12.6.3 Collaboration: Taxes, Tips, and Totals

Work through each situation with your partner. Be sure to show your work.

1. Xavier visits an Italian restaurant for dinner.

- a. If Xavier tips 20% on the subtotal \$24.95, how much tip will he pay?
\$4.99

Date:	Sep 19th
Time:	6:04 PM
Server:	Flo
Bread Stix	9.50
Ravioli Bites	10.50
Cheesecake	4.95
Subtotal	24.95
Sales Tax	_____
Tip	_____
Total	_____

- b. The city has a sales tax rate of 9.5%. How much in tax will the customer pay on the subtotal of \$24.95?
\$2.37
- c. What is the total Xavier spent on his meal, including tip and tax?
\$32.31



Think-Pair-Share

Ask students to initially complete the questions independently. Then have students discuss their work with a partner. These discussions will help students gain confidence and understanding.



Consider asking students the following:

- Does the sales tax amount depend on the tip amount? Or are those two separate, additional costs?
- Are these amounts related to the subtotal or the total cost?

12.6.3 Collaboration: Taxes, Tips, and Totals (continued)

2. A sporting goods store purchased baseball hats from a supplier for \$24 each and marked them up to \$35.
 - a. Calculate the percentage of the markup.
46%
 - b. A customer orders 4 baseball hats from the store online. What is the subtotal of the 4 hats?
\$140.00
 - c. If the sales tax is 6.25% of the subtotal and there is a \$7.00 shipping fee applied after tax, what is the total amount the customer paid?
\$155.75
3. Leilani found a 30% off discount code for an NXT True Hybrid Surfboard. The original cost of the surfboard is \$700. If the company does not charge tax but instead charges a \$150 shipping fee, how much does the surfboard cost in total?
\$640.00



Consider having students write their work on individual dry-erase boards to make sharing their work during the debrief easier.

12.6.4 Your Turn!

Component Overview: Students practice completing multi-step problems involving taxes, tips, fees, markups, and discounts.



12.6.4 Your Turn!

1. A car dealership pays a wholesale price of \$12,000 to purchase a vehicle. The car dealership wants to make a 32% profit.
 
 - a. How much money is added to the wholesale price to make a 32% profit?
\$3,840
 - b. After the markup, what is the total retail price of the vehicle?
\$15,840
 - c. The sales tax in the city it is sold is 7.5%. After tax, what will the customer pay for the car?
\$17,028
2. The table shows the pre-tax cost of items at a convenience store where the sales tax rate is 4.6%.

Item	Cost (\$)
Milk	3.75
Bag of Ice	2.50
Aspirin – 24 tablets	8.25
Snack Mix	3.75
Candy Bar	1.25
Batteries	6.75

- a. Cal stopped at the convenience store and picked up a bag of ice and batteries. What was his total cost, with tax?
\$9.68



Consider combining the 12.6.3 Collaboration and Your Turn! into stations and having students rotate through the problems.

12.6.4 Your Turn! (continued)

- b. Ms. Isales has \$10.00 in her wallet. Does she have enough to buy aspirin and a candy bar? Explain your reasoning using mathematics.

Yes, because when you add the items and find the tax it adds up to \$9.94 which is less than \$10.

3. A local restaurant automatically adds a 21% gratuity (tip), pre-tax, to each bill for parties over 10. The table shows the bills due for several parties over 10. Complete the table.

Party	Subtotal (\$)	Sales Tax (\$)	Gratuity (\$)	Total Amount of Bill
Flores Party	93.75	5.86	\$19.69	\$119.30
Atkinson Party	176.99	\$11.06	37.17	\$225.22
Mikelson Party	135.50	\$8.47	\$28.46	172.43



Consider asking students the following:

- What is gratuity? What does it mean that gratuity is pretax?
- How can you find the sales tax percentage from the sales tax and subtotal?
- Can you find the sales tax and gratuity separately?

12.6.5 Wrap-Up: Ticket Out the Door

Component Overview: Students solve a multi-step problem involving a discount, sales tax, and shipping fee. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should be allowed to access a calculator to complete the Wrap-Up.



12.6.5 Wrap-Up: Ticket Out the Door

1. Diesel has been waiting patiently for the Evolve Carbon All Terrain Skateboard to go on sale. For his birthday, he received a 25% off discount. The original price of the skateboard is \$1,999.99 and the company has a 5% sales tax, along with a flat fee of \$35 for shipping.

What is the total amount Diesel needs to purchase his dream skateboard?

\$1,609.99



Challenge students who finish early to write their own multi-step problem. Then have students exchange problems with a partner and solve each other's problem.

12.7 Percent Increase and Decrease

Lesson Introduction

 Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.		 Learning Target	I can find percent increase or percent decrease.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.		N/A	
 Essential Questions	How can you model percent change problems with a tape diagram and double number lines? How do you calculate percent increase and percent decrease?			
 Lesson Narrative	Students solve a variety of real-world percent change problems. Initially, students use tape diagrams and double number lines to model the problems. Then, students are introduced to the percent of change formula and apply that to additional problems in the lesson.			
 Component Summary	12.7.1 Warm-Up (~5 min.)	Use a percentage to determine the fewest number of people surveyed (<i>SE p. 247</i>)	12.7.4 Collaboration (10-15 min.)	Solve problems involving percent change (<i>SE p. 250</i>)
	12.7.2 Guided Instruction (10 min.)	Learn how to find the percent of change using models (<i>SE p. 248</i>)	12.7.5 Your Turn! (5 min.)	Additional practice solving problems involving percent change (<i>SE p. 252</i>)
	12.7.3 Try It (10 min.)	Use models to represent percent change (<i>SE p. 249</i>)	12.7.6 Wrap-Up (~5 min.)	Error analysis of percent change (<i>SE p. 253</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Percent of change <u>Additional Terms:</u> N/A		N/A	
			Additional Preparation	
			Consider creating and displaying an anchor chart with instructions, examples, and visuals for students to reference.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to understand why they can add the percent increase to 100% or subtract the percent decrease from 100%. - Students may reverse the order of the values in the percent of change formula.	- If students struggle with percent change formulas, it may help to walk them through the process to get to $\frac{\text{difference}}{\text{original}}$. This includes students recognizing that the percent of change will be positive for an increase and negative for a decrease. - Consider allowing students to use calculators for the 12.7.4 Collaboration, 12.7.5 Your Turn!, and 12.7.6 Wrap-Up. Allowing students to use calculators during the lesson will help students focus on discovery and the process of calculating percent of change. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers. For example, students should not need a calculator in the components prior to 12.7.4 Collaboration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns	MA.K12.MTR.4.1: Discussion and Reflection
	For the 12.7.4 Collaboration, have students take turns explaining how they solved questions 1 and 2. Partners can provide feedback and compare their work. These discussions will help students gain confidence and understanding.	The 12.7.6 Wrap-Up provides the opportunity for students to analyze the percent of change error in a sample of student work. Through identifying and correcting mistakes in a sample of student work, students will be less likely to make mistakes themselves.
Lesson Notes:		

12.7.1 Warm-Up: Interpreting Percentages

Component Overview: Students interpret percentages in a real-world problem by considering what the fewest number of people surveyed was.



12.7.1 Warm-Up: Interpreting Percentages

1. If exactly 93.6% of people completed a survey, what is the fewest number of people that were surveyed? Consider: How could it be determined if only one person could have completed it? Can partial people complete the survey?

125 is the fewest number of people that could have been surveyed. If one person completed the survey, then the percentage would have to be 100%. You cannot have a fractional part of a person complete a survey. For exactly 93.6% of people to complete a survey, the amount of people surveyed must be at least 125.



This component is scaffolded such that parts could be released to students at this point in the unit. Consider which questions may need more modeling or elaboration and which would be best for students to complete with a partner.

2. Explain your answer using mathematics.

I rewrote the percentage as a fraction and then used equivalent fractions to eliminate the decimal by putting the percentage over 1000 since I can not have a decimal amount of a person. Then, I reduced it to lowest terms.

$$\frac{93.6}{100} \times \frac{10}{10} = \frac{936}{1000} \div \frac{8}{8} = \frac{117}{125}$$

12.7.2 Guided Instruction: Percent of Change

Component Overview: Students develop an understanding of percent change by modeling problems with tape diagrams and double number lines. The models provide students with visual entry points to the concept of percent change.



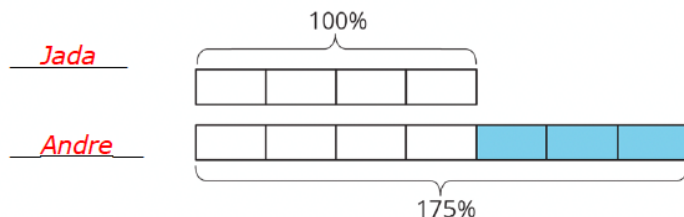
12.7.2 Guided Instruction: Percent of Change

Percentages can be used to make comparisons from one quantity to another.

- It takes Andre $\frac{3}{4}$ times longer than it took Jada to get to school.

The diagram models this in terms of percentages.

- Label each line of the tape diagram with the name of the student who is represented.



- Complete the statements.

The time it takes Andre to get to school is 75%

- ☐ more
☐ less

than the time it takes Jada. The diagram shows Andre's

travel time is

- ☐ 75%
☐ 175%

of Jada's travel time.

- Let x represent Jada's time. Write an expression to model how long it takes Andre to get to school.
 $1.75x$
- If it takes Jada 8 minutes to get to school, calculate how long it takes Andre to get to school.
 14 minutes



Consider asking students the following:

- Does it take Andre more or less time to get to school than Jada?
- What percentage is equivalent to $\frac{3}{4}$?

12.7.2 Guided Instruction: Percent of Change (continued)

The terms “percent increase” and “percent decrease” are used to describe a specific **percent of change**.



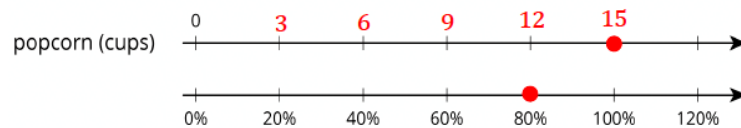
A **percent of change** is the difference between a final value and an initial value, expressed as a percentage of the initial value.



If time allows, consider having students share different strategies they used, or describe where they got stuck if they weren't able to determine an answer.

A double number line can also be used to model percent increase or percent decrease.

2. A movie theater decreased the size of popcorn bags by 20%. The old bags held 15 cups of popcorn.



- On the top number line, plot the 15 cups of popcorn as the initial value of 100%.
Shown on top number line.
- On the second line, plot 20% less than the initial 100%.
Shown on second number line.
- Use the double number line to find the number of cups in the smaller bag.
12 cups
- Discuss with your partner how the double number line is helpful for determining the impact of a 20% decrease in the size of the popcorn bag.
Answers will vary. Each segment on the number line represents 3 cups which is equivalent to 20%. A decrease of 20% shows that 3 cups need to be taken away from the initial 15 cups.



If the amount of popcorn is decreased by 20%, what percentage of the original size is the new size?

12.7.3 Try It: Percent of Change

Component Overview: Students practice using tape diagrams and double number lines to model percent change problems. Consider having students discuss their work with a partner after initially completing the questions independently.



12.7.3 Try It: Percent of Change

1. Explain which diagram, A or B, matches each situation.

Diagram A	Diagram B
Last year 100% This year 25%	Last year 100% This year 25%



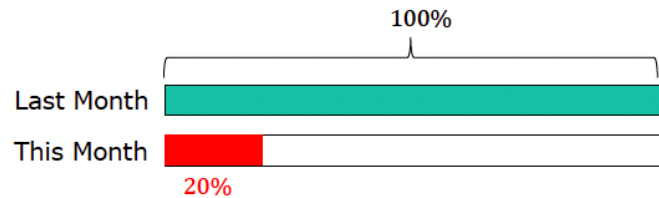
Ask students how they can distinguish between the two diagrams.

- How can you tell if the question pertains to percent increase based on the diagram?
- How can you tell if the question pertains to percent decrease based on the diagram?

Situation	Matching Diagram	Reason
Compared to last year's strawberry harvest, this year's increased 25%.	A	Diagram A matches because this year is 25% more than last year.
This year's blueberry harvest is 75% of last year's.	B	Diagram B matches because this year is 25% less than last year.
Compared to last year, this year's peach harvest decreased by 25%.	B	Diagram B matches because this year is 25% less than last year.
This year's plum harvest is 125% of last year's plum harvest.	A	Diagram A matches because this year is 25% more than last year.

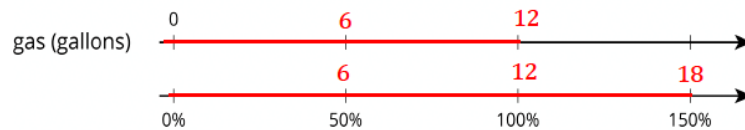
12.7.3 Try It: Percent of Change (continued)

2. The number of mosquitoes decreased by 80% from last month to this month. Create a tape diagram of this month to represent the situation.



If there were 100 mosquitoes last month, how many were there this month?

3. The gas tank in Robert's truck holds 12 gallons. The gas tank in his mom's truck holds 50% more gas. Using the double number lines, determine how many gallons the gas tank of Robert's mom's truck holds.



The gas tank in Robert's mom's truck holds 18 gallons.

12.7.4 Collaboration: Calculating Percent Increase or Decrease

Component Overview: Students collaboratively solve percent increase and decrease problems. Students are introduced to the percent change formula and then use it to solve a problem.



12.7.4 Collaboration: Calculating Percent Increase or Decrease

Percent increase and decrease can also be found through calculations.

Complete the following with your partner.

1. A new cereal box contains “20% More” cereal. The original cereal boxes contained 500 grams of cereal.



- a. Explain how to determine how much cereal is in the new box.
Find what 20% of 500 is. Then add it to 500. This will be 120% of 500.

- b. How much cereal comes in the new box?

$$500 + 0.2(500) = 600 \text{ grams}$$

2. The price of a shirt at ClothesMart is \$18.50, but there is a store coupon that lowers the price by 20%.

- a. Work with your partner to determine how much the shirt is with the coupon.

$$18.5 - 0.2(18.5) = \$14.80$$



Take Turns

Ask students to take turns solving each problem and then explaining their solution. This routine ensures each and every student has the opportunity to articulate their mathematical thinking and critique the reasoning of others.



There are multiple strategies for solving the problems in this component. Encourage students to use strategies that make sense for them as they build fluency in answering questions 1 and 2.

12.7.4 Collaboration: Calculating Percent Increase or Decrease (continued)

The table shows two different ways to complete this problem, along with corresponding answers.

Approach 1	Approach 2
The cost of the shirt with the coupon will be 80% of the original cost.	The new cost of the shirt is what's remaining after the 20% has been deducted from the original cost.
New cost = $0.8(18.50)$	New cost = $18.50 - 0.2(18.50)$



Algebra Talk

Ask students to compare the numeric expressions and discuss what is the same and different. How is the 20% discount represented in each?

- b. Explain whether or not these equations are equivalent.

These equations are equivalent because both can be used to find the new cost correctly. In Approach 1, the discount is removed from the percentage to find 80% of the original cost. In Approach 2, the price of the discount is first calculated and then subtracted from the original cost.

- c. Did your answer to Part A match the solution(s) of the equations in the table?

Yes



$$\text{Percent of Change} = 100 \times \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$$

3. The original cost of pants at ClothesMart is \$30.00, and after a store coupon is applied, the cost is only \$25.50.

- a. Determine the percent of change by filling in the blanks of the percent of change ratio for the situation.

$$\frac{\boxed{25.50} - \boxed{30}}{\text{Initial}} = \frac{\boxed{25.50} - \boxed{30}}{\boxed{30}} = \frac{\boxed{-4.5}}{30} = \boxed{-15} \%$$

12.7.4 Collaboration: Calculating Percent Increase or Decrease (continued)

- b. Explain whether the percent of change represents a percent increase or decrease based on its value.

This is a percent decrease because a discount decreases the price. You can also tell because the resulting change is a negative percentage.

- c. Complete the statements.

When calculating the percent of change using the ratio,

a positive value represents a percent

- ☐ increase,
☐ decrease,

and a negative value represents a percent

- ☐ increase.
☐ decrease.

In this context, it makes sense that the percent change is a

percent

- ☐ increase
☐ decrease

because the change is the result of a

discount.



Students may struggle to understand the series of boxes in question 3. If needed, walk through the steps with students or draw the problem on the board. Label each box with the information it should contain such as “initial cost,” “cost after coupon,” “difference,” and “percent change.” Then, have students fill the numbers in the boxes and use them to verbally explain the problem.

Be sure to provide students the opportunity to try it with their partner before jumping in to model. Consider also sharing strategies you observe on the board and asking students to compare them and determine errors.

12.7.5 Your Turn!

Component Overview: Students practice solving problems involving percent change.



12.7.5 Your Turn!

1. A school had 1,200 students last year and only 1,080 students this year. What was the percentage decrease in the number of students?

$$\frac{1080 - 1200}{1200} = \frac{-120}{1200} = -10\%$$

There was a 10% decrease in the number of students.



Consider connecting the context in these and other problems in the lesson to familiar or local situations to help students make connections to the problems. For example, perhaps there was a change in the number of students at their school or there is a package advertising 20% more that could be brought in for reference. Student motivation can be ignited if the problems involve people, places, and things that they are interested in.

2. A juice box has 20% more juice in its new packaging. The original packaging held 12 fluid ounces. How much juice does the new packaging hold?

$$1.2(12) = 14.4 \text{ fluid ounces}$$

12.7.6 Wrap-Up: Error Analysis

Component Overview: Students identify and correct an error in a percent change calculation. Consider using this information from this formative assessment to determine which students need remediation before the next lesson.



12.7.6 Wrap-Up: Error Analysis

- The number of campers who attend a local summer camp has increased from 1,938 to 2,128. The camp director calculates the percent of increase for an upcoming presentation. Her work is shown.

$$\begin{aligned} 2,128 - 1,938 &= 190 \\ 190 \div 2,128 &= 0.089 \\ 0.089 &= 8.9\% \end{aligned}$$

Explain the camp director's mistake.

The answer should be about 9.8% because $\frac{2,128-1,938}{1,938}$ is about 0.098. The camp director divided the change in number of campers by the new number instead of the original.

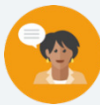


Students who are done early can also be asked to find the correct answer.

12.8 Calculating Percent Error and Simple Interest

Lesson Introduction

 Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.		 Learning Targets	- I can find percent error. - I can solve problems that involve simple interest.	
 Benchmark Coherence	Building On		Working Toward		
	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.		N/A		
 Essential Questions	- What is percent error? - How do you calculate percent error?				
 Lesson Narrative	After the 12.8.1 Warm-Up, students begin with an Exploration where they compare the actual versus estimated amounts in different contexts using a variety of strategies learned throughout the unit. This leads into introducing them to the concept of percent error, how to calculate it, and how to use it to solve real-world problems. Students then work with their peers and then independently to practice applying this in context. The lesson culminates with students reflecting on unit learning targets.				
 Component Summary	12.8.1 Warm-Up (~5 mins.)	Determine the better deal based on discounts (<i>SE p. 254</i>)	12.8.4 Guided Practice (8-10 mins.)	Practice using percent error to solve real-world problems (<i>SE p. 256</i>)	
	12.8.2 Exploration (7-10 mins.)	Solve real-world percent increase and decrease problems (<i>SE p. 254</i>)	12.8.5 Guided Instruction (10-15 mins.)	Develop an understanding of simple interest (<i>SE p. 257</i>)	
	12.8.3 Guided Instruction (10 mins.)	Learn how to find percent error (<i>SE p. 255</i>)	12.8.6 Wrap-up (~5 mins.)	Reflect on understanding of unit learning targets (<i>SE p. 259</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Percent error - Simple interest <u>Additional Terms:</u> N/A		N/A		Consider sharing the focus of today's lesson with grade-level science teachers, as the teachers may be able to help you make connections to how scientists use percent error.

 Teacher Tips	Common Misconceptions - Students may struggle to differentiate between the actual value and the intended value when finding the percent error. - Students may confuse the concepts of percent error and percent change. Use this as an opportunity to connect the concepts and highlight their differences. - Students may try to round answers when not prompted to do so.	Things to Consider - Encourage students to highlight and label the important values in the problem. When doing so, be sure that students are still focusing on the context and the meaning of the values in it by prompting them to discuss the contexts and the meaning of percent error in them. - Allowing students to use calculators during the lesson will help students focus on discovery and on the process of calculating percent of change. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers. - Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 12.8.4 Guided Practice, have students initially solve the problems independently. Then have students discuss their work. These discussions will help students gain confidence and understanding.	MA.K12.MTR.3.1: Mathematical Fluency In 12.8.5 Your Turn!, students practice solving percent error problems. As students practice, they will develop efficient and generalizable mathematical methods based on what they've been learning throughout the unit.
 Differentiate Instruction	Encourage students to use the Mark-Up strategy to make annotations as they read problems. Ensure these annotations go beyond simply circling numbers to include markings related to the meaning of the values in the context.	
Lesson Notes:		

12.8.1 Warm-Up: Would You Rather?

Component Overview: Students compare two options for ordering pizza and decide which they would rather take.



12.8.1 Warm-Up: Would You Rather?

It's "Multiple Monday" and the more pizzas purchased, the greater the discount applied. If one pizza is ordered, the discount is 10%. If two pizzas are ordered, the discount is 20%, and for three pizzas, the discount goes up to 30%.



1. Would you rather order 3 pizzas on separate checks, or order 3 pizzas on the same check and split the cost?
I would rather have 3 pizzas on the same check.
2. Justify your answer using mathematics.
If the pizzas were \$10, they would be \$9 each if ordered separately. If ordered together, each pizza would be \$7.



Ask students if they have seen discounts at their local pizzeria. Bring in sales flyers from pizzerias to show students the applications of the math they are learning.

12.8.2 Exploration: Estimate vs. Actual

Component Overview: Students explore situations where the recommended or estimated value is different from the actual value. This component sets students up to be introduced to percent error in the next component.



12.8.2 Exploration: Estimate vs. Actual

- Instructions to care for a plant say, "Water with $\frac{3}{4}$ cup of water daily." The plant has been getting 25% too much water. How much water has the plant been getting? Include units.

$$\frac{3}{4} \times \frac{5}{4} = \frac{15}{16} \text{ cups}$$

- The pressure on a bicycle tire is 63 psi. This is 5% higher than what the manual states is the correct pressure. What is the correct pressure? Include units.

$$1.05x = 63; x = 60 \text{ psi}$$

- The crowd at a sporting event is estimated to be 2,500 attendees. The exact attendance is 2,486 people. By what percentage is the estimated attendance off?

$$\frac{2,500 - 2,486}{2,486} = \frac{14}{2,486} = -0.0056 = -0.56\%$$

- Discuss with your partner what strategies you used to solve questions 1 – 3.

Answers will vary. For the first problem, I multiplied the recommended amount by a fraction equivalent to 125% to determine the amount of water the plant has been getting. For the second problem, I set up a percent equation and solved it to find the correct original pressure. For the third problem, I used the percent of change formula.



As students work through the problems, listen to students' discussions and elicit responses about the processes they are using to solve problems.

Monitor that students are not rounding inappropriately, as none of the questions say to round.

Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common, careless errors that many students make.



There are multiple strategies for solving the problems in this component. Encourage students to use strategies that make sense for them as they build fluency in solving problems.

12.8.3 Guided Instruction: Percent Error

Component Overview: Students learn what percent error is and how to calculate it. Students use the percent error formula to solve several problems. Students discuss why the percent error is expressed as a positive percentage.



12.8.3 Guided Instruction: Percent Error



Percent error is the difference between the estimated number and the actual number as a percentage of the actual value.

Percent error can be used to describe any situation where there is a correct value and an incorrect value, and to describe the relative difference between them.

1. A teacher estimates there are approximately 600 students at his school, but in actuality there are 625 students enrolled.

- a. What is the difference between the teacher's estimate and the actual number of students enrolled at the school?

25

- b. Using the definition of percent error, finish filling out the ratio of the situation.

$$\left| \frac{\text{Expected Value} - \text{Actual Value}}{\text{Actual Value}} \right| = \left| \frac{-25}{625} \right| = 0.\underline{04}$$

- c. The percent error in the teacher's estimate is 4 ____%.

- d. Discuss with your partner why the percent error is positive. Summarize your discussion.

Percent error is positive because the absolute value is taken after finding the difference of the two values.

2. The front of a cereal box lists the net weight of the product is 21 ounces (oz). After further measurements, it is determined there are actually 23 oz of cereal in the box.

- a. What is the difference, in ounces, between the weight listed on the box and the measured weight?

2 ounces

- b. Find the percent error. Show your work.

$$\left| \frac{21 - 23}{23} \right| \cdot 100 \approx 8.7\%$$



Consider reading the questions aloud. Encourage students to highlight the important values in the questions and label them as the “expected value” or “actual value.” This will make it easier for students to use the formula. Encourage students to explain what the percent error means in the context to ensure they are not simply circling values, but also grasping the context presented.



Prompt students' thinking with the following:

- What does “net weight” mean? Do you think every box of cereal has the exact weight printed on the package?
- Which amount is the expected value?
- Which amount is the actual value?
- What does the percent error mean in this context?
- How does the percent error relate to the percent of change?

12.8.4 Guided Practice: Percent Error

Component Overview: Students use percent error to solve real-world problems. Students may solve the problems independently or work with a partner.



12.8.4 Guided Practice: Percent Error

1. A store owner estimated she would have 125 customers in her store on its opening day. She underestimated the number of customers by 8%. How many customers visited the store on opening day?

$$\frac{x}{125} = \frac{8}{100}; x = 10$$

$$125 + 10 = 135 \text{ customers visited the store}$$

2. To be labeled a jumbo egg, the egg is supposed to weigh 2.5 ounces (oz). Alana buys a carton of jumbo eggs and measures one of the eggs as 2.4 oz. What is the percent error?

$$\frac{0.1}{2.4} \approx 0.0417 = 4.17\%$$

3. A student estimates there to be 25.7 milliliters (mL) of a solution in his beaker in science class. He underestimated the amount by 12.5%. What was the actual amount of solution in the beaker? Round to the nearest thousandth.

$$0.875x = 25.7$$

$$x = 29.371 \text{ mL}$$

4. Lia wanted to take advantage of the pizza special on "Multiple Monday." She estimates it will cost her \$28 for two large cheese pizzas after the discount is applied. The actual cost is \$28.80. What is the percent error? Round to the nearest whole percent.

$$\frac{0.80}{28.80} \approx 0.0278 \approx 3\%$$



Think-Pair-Share

Have students initially solve the problems independently. Then, have students discuss their work. These discussions will help students gain confidence and understanding.



Consider allowing students to select two of the questions to complete. Students will be able to thoughtfully and thoroughly solve the problems without feeling overwhelmed by their workload. They also will be more motivated to solve the problems they select because they will be connected to their interests.

12.8.5 Guided Instruction: Simple Interest

Component Overview: Students learn what simple interest is, how to calculate it, and how to use it to solve problems in context.



12.8.5 Guided Instruction: Simple Interest

1. Kristina received \$100 in a birthday card and wants to put the money into her savings account, where she will earn an additional 2% each year it remains in her account. This is known as **simple interest**.



Simple interest is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.



Ask students what they know about interest. List examples on the board. Explain that financial institutions charge interest for allowing a customer to borrow money with a loan and pays a customer interest for keeping their money in a savings account. Answer any questions students have about the concept of interest.

- a. How much is 2% of \$100?
\$2.00
 - b. How much money will be in Kristina's savings account after one year if she makes no additional deposits or withdrawals after putting in her birthday money?
\$102.00
 - c. Compare your answer with your partner and discuss how you determined the answer for Part B.
I knew that Kristina had \$100 already and then earned an additional \$2 which equals \$102 when added together.
2. Suppose an investment of \$100 earns \$3.50 interest in one year.
 - a. What investment amount will earn \$61.60 in two years at the same rate?

$$0.035x = \frac{61.60}{2}; x = \$880.00$$
Students could also set up a percent proportion.

12.8.5 Guided Instruction: Simple Interest (continued)

- b. Ask two other partner pairs to describe how they found the initial investment. Record their initial investment and write your summary of their process. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Initial Investment	My Summary of Their Reason	Initials
\$880.00	<i>Answers will vary. They divided \$61.60 by 2 since there was 2 years and then found what amount \$30.80 is 3.5% of.</i>	
\$880.00	<i>Answers will vary. They found what amount 3.5% is of \$61.60 and then divided by 2.</i>	

- c. Review the initial investments and reasons provided by other partner pairs. If necessary, fix any mistakes you and your partner made.



Simple Interest Formula

$$I = p \times r \times t$$

where I is the interest earned, p is the principal starting amount of money, r is the interest rate per year, and t is the length of the investment.

3. Tonio took out a six-year car loan for \$15,600, with a 3% simple interest rate over the life of the loan.
- How much interest, in dollars, will Tonio pay annually?
 $I = (15,600)(0.03)(1) = \468.00 per year
 - How much will Tonio pay in interest over the course of **six** years?
 $I = (15,600)(0.03)(6) = \$2,808.00$



Consider asking students the following:

- How did you determine the initial investment amount for the previous question?
- How does the method you used relate to the simple interest formula?
- How could you use the formula to determine an interest rate or the length of the investment?



Consider allowing students to “shop” for a car locally and use the advertised price and a current, realistic percentage rate to see how much interest they would have to pay over the life of a 5-year loan.

12.8.5 Guided Instruction: Simple Interest (continued)

- c. What is the total amount Tonio will pay back at the end of his six-year loan?

$$15,600 + 2,808 = \$18,408.00$$

- d. Explain why the total amount Tonio will have paid back by the end of his loan is more than the loan amount he borrowed.

He will pay back more because he has to pay back the amount he borrowed, plus the interest accrued on the loan.

12.8.6 Wrap-Up: Reflection

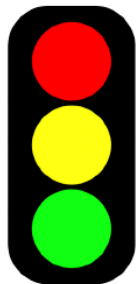
Component Overview: Students reflect on their understanding of the learning targets from the unit. Encourage students to provide specific details if time allows. Use observations from this reflection to determine where students need support before the summative unit assessment.

12.8.6 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about percentages and proportional relationships, including those listed below.

- Solving real-world problems using proportions
- Solving discount, markup, tax, tip, and fees problems
- Finding percentage of increase and decrease
- Finding percent error
- Solving problems involving simple interest

1. Use the stoplight reflection below to share how you are feeling about each of the concepts at this point.



I still feel stuck on how to ...

Answers will vary. There is no right or wrong answer, this is a student opinion.

I'm starting to get the hang of ...

Answers will vary. There is no right or wrong answer, this is a student opinion.

I feel confident and ready to go with ...

Answers will vary. There is no right or wrong answer, this is a student opinion.



Some students may struggle to express themselves in writing. Consider providing alternatives, such as using speech-to-text software or an audio recorder (i.e. Flip Grid), or by labeling examples.